Biology Grade 12 - Solved Exercises (BTBB)

Balochistan Textbook Board (BTBB)

Unit 14: Respiration

Exercise - Section I: Objective Questions (MCQs)

Q1. Pleura is a double layered thin membrane that covers:

- Heart
- Liver
- Kidney
- ✓ Lungs

Q2. Hemoglobin in man increases the oxygen carrying capacity of the blood to about: (Book text, p25: 'about 75 times'.)

- ✓ 75 times
- 50 times
- 60 times
- 100 times

Q3. Plasma proteins are involved in the release of CO2: (Book text, p29: 'Some CO2 (about 7%) transported ... with the help of plasma proteins'.)

- 70%
- ✓ 7%
- 30%
- 20%

Q4. Structure, which closes the passage to lungs when food is coming, is called:

Glottis

✓ Epiglottis

Uvula

Pharynx

Q5. Myoglobin loses oxygen at: (Book text, p25: 'gives oxygen when partial pressure of oxygen is below 20 mm of Hg'.)

60 mm Hg

19.6 mm Hg

90 mm Hg

✓ 20 mm Hg

Q6. Breathing is an example of:

Diffusion

Osmosis

✓ Ventilation

Cellular Respiration

Q7. The structure which prevents entry of food to wind pipe is called:

Glottis

✓ Epiglottis

Tongue

Soft palate

Q8. Which one of the following lacks cartilage? (Book text, p20: bronchioles 'lack cartilages'.)

Trachea

✓ Bronchioles

Bronchi

Larynx

Q9. The pleural fluid surround the:

Liver

Kidneys

Heart

✓ Lungs

Q10. The percentage of CO2 carried in the form of bicarbonate is: (Book text, p26: 'As bicarbonate ions (about 70%)'.)

85%

60%

✓ 70%

65%

Q11. Each molecule of myoglobin combines with one molecule of: (Book text, p25: 'bind only one molecule of oxygen'.)

✓ Oxygen

Carbon dioxide

Nitrogen

Sulphur

Q12. In human, respiratory pigment is:

Haemocyanin

Haemoerythrin

Chlorocrurin

✓ Haemoglobin

Q13. The residual volume of air in human lung is: (Book text, p23: 'About 1200 - 1500 ml air always remains in the lungs ... this is called residual volume'.)

- 2.5 liter
- 5.0 liter
- ✓ 1.5 liter
- 3.0 liter

Q14. Chemotherapy and radiotherapy may help in the treatment of: (Book text, p31: 'Chemotherapy and radiotherapy may treat cancer'.)

- Flu
- Emphysema
- ✓ Lung cancer
- Asthma

Q15. Mycobacterium tubercle causes: (Book text, p32: 'Pulmonary tuberculosis is caused by ... Mycobacterium tubercle'.)

- Emphysema
- Sinusitis
- Pneumonia
- ✓ Pulmonary tuberculosis

Exercise - Section I: Fill in the blanks

Respiratory surface must be permeable, so that ____ can pass through it.

respiratory gases

The pharynx is part of both the respiratory and ____ system.
digestive

The trachea divides into two primary ____.
bronchi

Each lung contains about 35 millions ____. (NOTE: unit text p20 says 'about 350 million alveoli'; the exercise prints '35 millions'.)
alveoli

About 97 - 98% of O2 is carried by the RBC as ____.
oxyhaemoglobin

The chloride shift is also called ____ phenomenon.
Hamburger's

Sinusitis is an inflammation of nasal ____.
sinuses

Haemoglobin can carry ____ oxygen.
four molecules of

____ media is an inflammation of middle ear.
Otitis

Mycoplasma cause lower respiratory infection named ____. (Book

text, p30: Mycoplasma is listed among the causes of pneumonia.)
pneumonia

Exercise - Section II: Short Questions

Q1. Differentiate between internal respiration and external respiration.

External respiration is the exchange of respiratory gases (O2 and CO2) between the organism and its environment, and between the circulatory fluid and the external environment; no ATP or water is formed, and it is aerobic. Internal respiration is a biochemical process within the cell that oxidizes food molecules; it is the exchange of gases between the cell and the circulatory fluid, and ATP, CO2 and water are produced. It may be aerobic or anaerobic.

Q2. Write three properties of respiratory surfaces.

A respiratory surface shows three characteristics for ready exchange of gases through diffusion: (1) it should be large, moist and highly permeable for easy exchange of gases; (2) it should be a thin epithelium (1 mm or less) and possess a network of blood capillaries, which facilitates rapid transport and diffusion of gases between the surface and blood; (3) a good ventilation mechanism should be present to maintain a steep diffusion gradient.

Q3. Describe briefly about the functions of nasal cavities.

The nasal cavities are the two vestibules separated by the nasal septum. They contain a network of hair and are lined by mucus membranes (which secrete mucus) along with cilia; these serve as a defence mechanism, trapping dust and solid particulate

substances present in the air, which the cilia push to the pharynx for removal. The mucus also moistens the air and brings the temperature of inhaled air close to body temperature (about 30 degrees C depending on external temperature).

Q4. Define Bronchi, bronchioles and alveolar ducts.

Bronchi: the trachea first divides into two primary bronchi, one entering each lung, where each bronchus redivides into many bronchi made of plate-like cartilages. Bronchioles: the bronchi finally give rise to smaller membranous tubes less than 1 cm in diameter, which lack cartilages and are made of ciliated cuboidal epithelium and a layer of circular smooth muscles. Alveolar ducts: the bronchioles divide and subdivide to form respiratory bronchioles which eventually terminate into alveolar ducts; these ducts open into small collections of air sacs known as alveoli.

Q5. Write any three differences between inspiration and expiration.

Inspiration: the muscles of the ribs contract so the ribs move upward and outward, and the diaphragm moves downward; the chest cavity increases, its pressure drops below atmospheric pressure, and air rushes through the respiratory passage into the lungs. Expiration: the volume of the chest cavity is decreased as the rib muscles relax and the ribs move downward and inward, while the diaphragm muscles relax and the diaphragm projects into the thoracic cavity becoming dome-shaped; the elastic lungs contract and force the air out, so the air inside the lungs moves out. Tidal volume is the amount of air moved in and out with each quiet breath, normally 500 ml, while inspiratory reserve volume is

the extra 3000 to 4000 ml of air that can be inspired during forced inspiration.

Q6. Difference between tidal volume and inspiratory reserve volume.

Tidal volume is the amount of air moved in and out with each quiet breath, which is normally 500 ml. Inspiratory reserve volume is the additional amount of air, about 3000 to 4000 ml, that can be inspired by forced inspiration over and above the tidal volume. (Book text, p23.)

Q7. Write short note on myoglobin.

Myoglobin is another respiratory pigment in man, present in the muscles of humans and other mammals (which is why meat is red). It is made of one iron-containing polypeptide chain of 154 amino acids and binds only one molecule of oxygen. Its function is to store oxygen in the muscle and give it up when the partial pressure of oxygen falls below 20 mm of Hg; it has a greater affinity for oxygen than haemoglobin.

Q8. Describe transport of CO2 as carboxyhaemoglobin.

About 23% of the carbon dioxide is transported as carboxyhaemoglobin: CO2 combines with the globin part of haemoglobin in the red blood cells to form carboxyhaemoglobin (carbaminohaemoglobin). This compound is carried in the blood to the lungs, where, because the partial pressure of CO2 is less than in blood, it breaks again and releases CO2, which diffuses out of the capillaries into the alveoli of the lungs and is expelled by expiration.

Q9. Write causes and symptoms of sinusitis.

Sinusitis is the inflammation of the sinuses, the holes in the skull between the facial bones, which are lined by mucus-secreting membrane. Causes: it is generally caused by atmospheric pollution, dust, smoke, cold and wet climate, excessive dryness, and bacterial or viral infections. Symptoms: the inflamed, mucus-filled sinuses cause pain and pressure in the face and a blocked nose, and the condition may be acute (symptoms lasting 2-8 weeks) or chronic (slowly progressing, lasting much longer). Treatment: antibiotics or sulfa drugs for bacterial infection, antiallergic drugs and decongestants, and steam inhalation called nebulization.

Q10. Describe the treatment of lung cancer.

Lung cancer is a malignant tumour, one of the most common cancers in the world, caused by smoking and inhalation of unhealthy air; about 90% of lung cancer is caused by smoking, and the chances are ten times higher in smokers or people living in crowded smoky areas. Treatment: the tumour or the whole lung is removed completely by operation (pneumonectomy) before secondary growth occurs, and chemotherapy (chemical agents) and radiotherapy (ionizing radiation) may also treat or control the cancer.

Exercise - Section III: Extensive Questions

Q1. Explain breathing mechanism in humans.

Breathing (ventilation) takes place in two phases. Inspiration: the rib muscles contract, the ribs move upward and outward, and the

diaphragm moves downward; the chest cavity increases and its pressure drops below atmospheric pressure, so air rushes through the respiratory passage into the lungs. Expiration: the rib muscles relax, the ribs move downward and inward, the diaphragm relaxes and becomes dome-shaped, projecting into the thoracic cavity; the chest cavity is reduced from the sides and the floor, the elastic lungs contract, and the air inside the lungs is forced out. This rhythmic change in chest volume, driven by the ribs and diaphragm, is the breathing mechanism in humans.

Q2. Describe respiratory tract up to bronchi.

The respiratory tract (air passageway) consists of the external nostrils, the nasal sacs or cavities, the internal nares, the pharynx, the larynx, the trachea and the bronchi. The nose is the only externally visible part, made of bones, cartilages and fatty muscle tissue; its cavities trap dust by hair, mucus and cilia and moisten and warm the air. The pharynx is a muscular, mucus-secreting passage connecting the oral and nasal cavities to the oesophagus and larynx, with three sections (nasopharynx, oropharynx, laryngopharynx). The larynx (voice box) is made of cartilages and muscles; its epiglottis covers the glottis during swallowing, and its vocal cords produce voice. The trachea is a ciliated membranous windpipe, about 10-12 cm long, with 15-22 C-shaped cartilaginous rings that keep it open; it runs ventral to the oesophagus up to the 5th thoracic vertebra and divides into two primary bronchi.

Q3. Explain causes, symptoms and treatment of tuberculosis.

Tuberculosis (pulmonary tuberculosis) is an infectious bacterial

disease of the lower respiratory system, more common in poor people due to poor living conditions and malnutrition. Cause: the bacillus bacterium Mycobacterium tubercle (about 15% develop TB of lymph nodes, joints and the gastrointestinal tract). Symptoms: the inside of the lungs is damaged, alveoli burst and are replaced by inelastic connective tissue; the lung cells form a protective capsule (tubercle, a small rounded swelling) around the bacteria; the patient has cough, fever, pale face and sweating at night, and in severe form chest pain and breathlessness. Treatment: the disease is curable with appropriate drug therapy such as antibodies for 9 months regularly (Daily Observed Treatment Short Course, DOTS); vaccination is also available, and because it is contagious the patient should be kept in isolation to prevent infection.

Q4. Write a detail note on transport of CO2 in human.

Carbon dioxide from the tissues passes into the blood plasma in the tissue capillaries, where its transport occurs by three ways. (1) As bicarbonate ions (about 70%): CO2 combines with water inside the red blood cells, in the presence of carbonic anhydrase, to form carbonic acid, which splits into hydrogen ions and bicarbonate ions; the bicarbonates diffuse out of the red blood cells and are carried by the plasma, while the hydrogen ions are absorbed by the globin protein of haemoglobin. At the lung capillaries the bicarbonate ions recombine with hydrogen ions to form carbonic acid, which splits into water and CO2 that diffuses into the alveoli and is expired. (2) As carboxyhaemoglobin (about 23%): CO2 combines with haemoglobin and is released again at the lungs. (3) As dissolved CO2 in plasma (about 7%), and a further small part with the help of

plasma proteins, which is a rather inefficient way to carry CO2.

Q5. Write a detail note on transport of Oxygen in human.

About 97-98% of oxygen is carried by haemoglobin, while the remaining 2-3% is transported as dissolved oxygen in plasma. Haemoglobin increases the oxygen carrying capacity of the blood in humans to about 75 times: iron in haemoglobin combines loosely with oxygen in the red blood cells of the pulmonary capillaries to form oxyhaemoglobin (Hb + 4O2 gives 4O2Hb), and this bright red blood circulates to the tissue cells, where oxyhaemoglobin releases oxygen because haemoglobin gives up its oxygen where the oxygen concentration is low, and the blood becomes purple red. The maximum oxygen carrying capacity of blood at sea level is 20 ml per 100 ml of blood (100% saturation); haemoglobin loses oxygen when the partial pressure falls below 60 mm of Hg. Myoglobin in the muscles stores oxygen and releases it when the partial pressure falls below 20 mm of Hg.

Q6. Explain causes, symptoms and treatment of emphysema.

Emphysema is a lung disorder in which the air sacs (alveoli) degenerate and the elastic fibres in them are destroyed, so the alveolar walls degenerate and small alveoli combine into larger ones; this gives fewer alveoli with increased volume and decreased surface area for gaseous exchange. Causes: the root cause is long-term irritation of the lungs by cigarette smoke, polluted air, industrial dust, or exposure to certain drugs and coal; substances in tobacco smoke weaken the alveolar walls. Symptoms: increasing breathlessness, difficulty in walking, loss of lung elasticity so

exhalation becomes difficult and much air remains in the lung during expiration, inflammation and narrowing of bronchioles, fatigue, coughing and cyanosis (blue skin). Treatment: there is no cure, but care may help — avoid smoking, treat all respiratory infections immediately, and use oxygen equipment and respiratory devices.

Q7. Illustrate the effects of smoking on human health.

Smoking has many effects on the respiratory system and human health. The book states that 87% of cigarette smokers develop lung cancer, and smoking also causes cancer of the mouth, larynx and oesophagus, as well as chronic bronchitis and emphysema. Cigarette smoke contains chemicals that irritate the respiratory tract and lungs, causing early-morning coughing and wheezing; it damages or destroys the cilia, so microbes cannot be trapped and easily settle in the respiratory system, making smoking an indirect cause of pneumonia. Smokers' lungs appear dark and mottled, unlike the light pink lungs of healthy people.

Q8. Write note on otitis media.

Otitis media is an inflammation of the middle ear in which the Eustachian tube (the tube between the middle ear and pharynx) fills with fluid and becomes closed. Causes: infection, allergy, recurrent attacks of common cold, blockage of the Eustachian tube, nutritional deficiency, sinusitis and measles. Symptoms: sudden and severe earache, deafness, fever, headache, a sense of fullness of the ear, tinnitus (ringing or buzzing in the ear), fluid leaking from the ear, and difficulty in speaking and hearing; sometimes the

eardrum bursts, discharging pus and relieving pain. Treatment: in most cases (around 80% of patients) the fluid clears within three to four days and the eardrum repairs itself; for complicated cases antibiotic therapy is prescribed, and painkillers may be given to relieve pain and fever.

Q9. Explain alveolar duct and alveoli.

The bronchioles divide and subdivide to form respiratory bronchioles, which eventually terminate into alveolar ducts. These ducts open into small collections of air sacs known as alveoli, the actual sites of gaseous exchange. The alveoli are made of simple squamous epithelium only 0.1 cm thick (as printed) and have a rich network of capillaries, so gases diffuse rapidly between air and blood. Each lung possesses about 350 million alveoli, giving an enormous surface area for gaseous exchange; the alveoli also contain collagen and elastin proteins which allow them to expand and recoil easily during breathing.

Q10. Write a note on lung cancer.

Lung cancer is a malignant tumour (uncontrolled cell division) and one of the most common cancers in the world. Causes: smoking and inhalation of unhealthy air; the chances are ten times higher in smokers or people living in crowded smoky areas, and about 90% of lung cancer is caused by smoking. Symptoms: in the initial stages the cells lining the bronchi thicken and harden (callusing) and the cilia of the nasal passage are destroyed; the tumour of disordered cells spreads and penetrates other tissues (metastases) and grows until the bronchus is blocked, cutting off the air supply to

the lung, while trapped secretions become infected (pneumonia or lung abscess). Treatment: surgical removal of the tumour or the whole lung (pneumonectomy) before secondary growth occurs, together with chemotherapy and radiotherapy to control or kill the malignant cells.

Unit 15: Homeostasis

Exercise - Section I: Objective Questions (MCQs)

Q1. Body temperature regulation in human is based on complex homeostatic thermostat present in:

- Thalamus
- Pons
- ✓ Hypothalamus
- Amygdala

Q2. Trimethylamine oxide is produced in:

- ✓ Cartilaginous / marine fish
- Bony fish
- Jawless fishes
- Lung fishes

Q3. Which of the following structure of kidney is responsible for the production of concentrated urine?

- Cortical nephrons
- ✓ Juxtamedullary nephron
- Vasa recta
- Glomerulus

Q4. Animals that generate their own body heat through heat production as by product of metabolism:

- ✓ Endotherms

Ectotherms

Heterotherms

All of these

Q5. The hormone affects ascending limb: (NOTE: the book's hormonal-control text, p52, does not name the ascending limb explicitly; ADH is taken as the answer because ADH drives the countercurrent concentration mechanism described on pp51-52, in which Na+ and Cl- are transported from the ascending limb.)

✓ ADH

Aldosterone

Thyroxin

Both a and b

Q6. Concentrated urine is produced in mammals when there is:

Excess of water in the body

✓ Shortage of water in the body

Less urea in blood

Less glucose in blood

Q7. Water is mostly absorbed by: (NOTE: options (c) and (d) are both printed as '(c)' on the page; read as (c) Loop of Henle, (d) Collecting duct.)

✓ Proximal tubule

Distal tubule

Loop of Henle

Collecting duct

Q8. Glomerulus are located in: (Book Table 15.1: glomerular capillaries are 'Located in the mouth of the Bowman's capsule'.)

- Collecting duct
- Renal tubule
- Pelvis
- ✓ Renal capsule

Q9. The ideal or normal value of variable is called: (Book text, p40: 'This normal value is called set point'.)

- ✓ Set point
- Output
- Response
- Stimuli

Q10. The salt secreting glands in some marine fishes are called:

- Sebaceous glands
- Sweat glands
- ✓ Rectal glands
- Tear glands

Q11. The chief nitrogenous waste of fresh water animals is:

- Urea
- ✓ Ammonia
- Uric acid
- Creatinine

Q12. The useful substances are reabsorbed from:

- ✓ Renal tubule

Renal capsule

Collecting duct

Vasa recta

Q13. Ureters are less affected as compare to other urinary tract infections due to: (Book text, p53: 'Ureters are less infected as compare to other parts of urinary tract due to its acidic nature'.)

Large surface area

Less acidic

Narrow tube

✓ More acidic

Q14. The drug given to prevent kidney rejection is:

✓ Cyclosporine

Penicillin

Pain killers

Antibiotics

Exercise - Section I: Fill in the blanks

The maintenance of internal environment at constant state is called ____.

homeostasis

The term homeostasis was introduced by ____. (Book text, p40: 'coined by physiologist Walter Cannon in 1932 in his famous book Wisdom of the body'.)

Walter Cannon

Those animals that can maintain their osmotic conditions at constant state are called ____.

osmoregulators

Marine fish usually have ____ to remove their extra salts.

rectal glands

Animals which excrete urea as their chief nitrogenous waste are called ____.

ureotelic

The network of capillaries present in the mouth of Bowman's capsule is called ____.

glomerulus

The process of formation of concentrated urine is called ____ mechanism.

counter-current

Peritoneal dialysis uses the ____ cavity of body.

peritoneal

Hyperthermia is ____ body temperature.

increased

The cyclosporine is used to prevent ____.

rejection

Exercise - Section II: Short Questions

Q1. Define homeostasis and write its components.

Homeostasis is the ability of an organism to maintain a relatively stable internal state that persists despite changes in the outside environment. Its three components are: (i) Osmoregulation — maintenance of water and dissolved solutes in a nearly constant state; (ii) Excretion — removal of metabolic wastes from the body, especially nitrogenous wastes; (iii) Thermoregulation — maintenance of constant body temperature. Its mechanism has three parts: a receptor that detects the change (stimulus), a control centre that compares it with the set point, and an effector that carries out the response restoring the set point.

Q2. Write the importance of negative feedback mechanism.

Negative feedback is important because it keeps internal variables at their set point: when a variable deviates, the response reverses the deviation and restores the normal value. For example, when body temperature rises, sweating and vasodilation cool the body; when water is short, ADH increases water reabsorption. Without negative feedback, small deviations would grow into dangerous disturbances of the internal environment.

Q3. Write the importance of positive feedback mechanism for our body.

Positive feedback is important where a process must be driven to completion rather than reversed: the response amplifies the original change until a definite end point is reached. Examples in the body are the contractions of childbirth, which are strengthened

by oxytocin until delivery, and blood clotting, where each step accelerates the next until the clot forms. Such mechanisms are essential for processes that must finish quickly and completely.

Q4. Differentiate between osmoregulator and osmoconformer.

Osmoregulators are animals that maintain their internal osmotic conditions at a constant state regardless of the external environment; most vertebrates, including freshwater and marine bony fishes, are osmoregulators and spend energy to do so. Osmoconformers are animals whose internal osmotic concentration changes with (conforms to) the surrounding water; most marine invertebrates are osmoconformers and do not regulate actively.

Q5. Describe the problems faced by osmoregulators.

Osmoregulators face opposite problems in different habitats. In fresh water, water continuously enters the body by osmosis and salts are lost, so they must excrete large volumes of dilute urine and actively take up salts. In sea water, water is lost from the body and excess salts enter, so they must drink sea water and excrete the extra salts through rectal glands or chloride cells. On land, the main problem is water loss by evaporation, so they must conserve water by producing concentrated urine and reducing loss from body surfaces.

Q6. Write the importance of urea as chief nitrogenous waste of most terrestrial animals.

Urea is important as the chief nitrogenous waste of most terrestrial animals because it is far less toxic than ammonia, so it can be

stored and concentrated in the body without harm, and it is soluble in water yet requires much less water for excretion than ammonia does. This lets land animals conserve water while still removing nitrogenous waste safely — a key adaptation to life on land, where water is scarce.

Q7. Illustrate the two types of nephrons.

There are two types of nephrons. Cortical nephrons have short loops of Henle that lie mostly in the cortex and are the common type. Juxtamedullary nephrons have long loops of Henle that extend deep into the medulla, together with the vasa recta (long capillary loops); they are responsible for producing concentrated urine through the countercurrent mechanism. Both types have the same basic parts — Bowman's capsule with glomerulus, proximal tubule, loop of Henle, distal tubule and collecting duct.

Q8. Write the common causes of urinary tract infections.

The most common cause of urinary tract infection (UTI) is the bacterium Escherichia coli; other bacteria, viruses and fungi may also cause UTI. Sexual contact with infected partners and unhygienic conditions are the major causes of UTI. Infections are more virulent in females than males because of the shorter urethra in women, incomplete emptying of the bladder, and the large exposed surface area of the external genitalia. Symptoms include burning during urination, cloudy, dark or bloody urine, tiredness, fever or chill.

Q9. Describe the three groups of animals on the basis of heat

production.

On the basis of heat production, animals fall into three groups: (1) Ectotherms, which obtain most of their body heat from the environment; (2) Endotherms, which generate their own body heat as a by-product of metabolism; (3) Heterotherms, which show both patterns at different times. On the basis of the ability to thermoregulate, animals are also grouped as poikilotherms, which cannot maintain a constant internal temperature, and homeotherms, which maintain a nearly constant body temperature.

Q10. Define Lithotripsy briefly describe laser lithotripsy.

Lithotripsy is a medical procedure involving the physical destruction of hardened masses like kidney stones, bezoars or gallstones; the term comes from Greek words meaning 'breaking (or pulverizing) stones'. Its two types are: (1) Extracorporeal Shock Wave lithotripsy, which uses shock waves to break the kidney stones into small pieces that pass through the urinary tract and out of the body; the treatment usually takes 45 minutes to one hour and may be done under general anaesthesia so the patient feels no pain. (2) Laser lithotripsy, another treatment by ureteroscopy, in which a long tube-shaped tool is used to find, remove and break the stone; a laser breaks the stone into small pieces so they may pass through the urinary tract.

Q11. Write the problems associated with kidney transplant.

The problems associated with kidney transplant are: rejection of the donor kidney by the recipient's immune system, which is controlled with the drug cyclosporine; the shortage of suitable

donor kidneys and the need for blood and tissue matching; the risks of surgery itself; and the side effects of lifelong immunosuppressant drugs, which weaken immunity and increase the risk of infections. After transplant the patient must be monitored continuously to detect rejection or infection early.

Exercise - Section III: Extensive Questions

Q1. Describe relationship between excretory products and habitats.

The form of nitrogenous waste is related to the habitat and water availability. Animals in water (most aquatic animals, including freshwater fishes) excrete ammonia, which is highly toxic but diffuses out easily because water is plentiful. Terrestrial animals, which must conserve water, excrete urea: it is less toxic than ammonia and needs less water for removal, so mammals and most land vertebrates are ureotelic. Animals in very dry conditions (birds, reptiles, insects) excrete uric acid as semi-solid pellets, which needs almost no water but costs more energy to produce. Marine fishes face salt gain and water loss and use rectal glands to excrete excess salts; cartilaginous marine fishes also retain trimethylamine oxide (TMAO) and urea to match the osmolarity of sea water.

Q2. Define and explain structure of nephron.

A nephron is the structural and functional unit of the kidney. It begins with the renal corpuscle: a tuft of capillaries called the glomerulus, located in the mouth of the double-walled Bowman's capsule. The capsule leads into the proximal convoluted tubule, then the loop of Henle with descending and ascending limbs, then the distal convoluted tubule, and finally the collecting duct. Two

capillary networks serve it: the glomerular capillaries at the capsule and the peritubular capillaries around the tubule (in juxtamedullary nephrons these form the long vasa recta). Cortical nephrons have short loops in the cortex; juxtamedullary nephrons have long loops reaching deep into the medulla and produce concentrated urine.

Q3. Explain the steps of urine formation.

Urine formation takes place in three steps. (i) Glomerular filtration: blood pressure in the glomerulus forces water, salts, glucose, urea and other small substances out of the blood into Bowman's capsule, forming the glomerular filtrate, while blood cells and plasma proteins remain in the blood. (ii) Selective tubular reabsorption: as the filtrate passes along the renal tubule, useful substances — most of the water, glucose, amino acids and mineral ions — are reabsorbed into the peritubular capillaries; water is mostly absorbed by the proximal tubule. (iii) Tubular secretion: hydrogen ions, creatinine and drugs are removed from the blood through the peritubular capillaries into the renal tubule and collecting duct. The end product is urine. When water is short, the countercurrent mechanism concentrates the urine: Na+ and Cl- are transported from the ascending limb of Henle's loop by the vasa recta, building a concentration gradient from 300 mOsm in the cortex to 1200 mOsm in the inner medulla, and ADH increases water reabsorption from the collecting duct.

Q4. Define dialysis and explain the two types of dialysis.

Dialysis is the temporary treatment of kidney failure patients in which wastes are removed from the body through a dialyzing fluid,

because the kidneys are not working well enough for survival without it. Its two types are: (1) Haemodialysis, in which the patient's blood is passed through a dialyzer (artificial kidney) that removes wastes and excess water before the blood returns to the body. (2) Peritoneal dialysis, which uses the peritoneal cavity of the body itself as the filter: dialyzing fluid is introduced into the abdomen, wastes diffuse into it across the peritoneal membrane, and the fluid is then drained out.

Q5. Explain the kidney transplant process in detail.

Kidney transplant is a surgical procedure of placing a healthy kidney into the belly (abdomen) of a chronic kidney failure patient. A suitable donor kidney (from a living or deceased donor) is matched to the recipient by blood group and tissue type. The surgeon connects the donor kidney's blood vessels to the recipient's vessels and its ureter to the bladder, so the new kidney filters blood and forms urine. The main problem is rejection by the recipient's immune system; the drug cyclosporine is given to prevent rejection, and the patient needs lifelong monitoring for rejection, infection and drug side effects.

Q6. Describe the osmoregulatory adaptations in different habitats.

In fresh water, animals gain excess water and lose salts: they excrete large volumes of dilute urine and actively absorb salts through chloride cells in the gills. In sea water, animals lose water and gain salts: marine bony fishes drink sea water, excrete excess salts through rectal glands (and chloride cells), and produce small volumes of concentrated urine; cartilaginous fishes retain urea and

trimethylamine oxide to keep their blood iso-osmotic with sea water. On land, animals conserve water by producing concentrated urine (countercurrent mechanism and ADH), excreting urea or uric acid instead of ammonia, and reducing evaporation through body coverings and behaviour.

Q7. Describe the external and internal structure of human kidney.
Externally, the human kidney is a bean-shaped, reddish-brown organ. Internally it has an outer cortex and an inner medulla containing renal pyramids separated by renal columns; the pyramids end in papillae that drain into minor and major calyces, which join the renal pelvis, from which the ureter leaves the kidney. The kidney is covered by a fibrous capsule and receives the renal artery, while the renal vein drains it. The functional units are about a million nephrons per kidney: each has a Bowman's capsule with a glomerulus, a proximal tubule, a loop of Henle, a distal tubule and a collecting duct, served by glomerular and peritubular capillaries.

Q8. Define kidney stones. Explain types of lithotripsy.
Kidney stones (nephrolithiasis) are hard mineralized materials that cling together and form pebbles within the kidney or any other part of the urinary tract such as the ureter or bladder; they are small, about 2-3 mm in diameter, and one to many in number. Calcium stones (calcium oxalate and calcium phosphate) are the most common; uric acid stones form when urine is often too acidic. Symptoms include pain while urinating, blood in urine, sharp pain in the back and lower abdomen, nausea and vomiting. Treatment depends on size, composition and location: small stones are treated

with medicine, while large and complex stones need surgery or lithotripsy. The two types of lithotripsy are extracorporeal shock wave lithotripsy, which uses shock waves to break stones into small pieces, and laser lithotripsy by ureteroscopy, in which a laser breaks the stone into pieces that pass through the urinary tract.

Q9. Define feedback mechanism, describe in detail negative and positive feedback mechanism.

A feedback mechanism is a control system in which the output (response) feeds back to influence the input, keeping a variable near its set point. Its parts are a receptor that detects a change in the variable, a control centre that compares the value with the set point, and an effector that carries out the corrective response. In negative feedback the response reverses the deviation: for example, when body temperature rises, sweating and widening of skin blood vessels cool the body back to the set point; when water is short, ADH increases water reabsorption. In positive feedback the response amplifies the deviation until an end point is reached: for example, during childbirth the contractions stimulate further oxytocin release, strengthening contractions until delivery, and in blood clotting each step accelerates the next until the clot is complete.

Q10. Explain classification of animals on the basis of temperature and also write thermoregulatory strategies in man.

Animals are classified by the source of body heat as ectotherms (heat from the environment, e.g. most fishes, amphibians, reptiles), endotherms (heat generated by their own metabolism, e.g. birds

and mammals) and heterotherms (showing both patterns at different times). By the ability to thermoregulate they are poikilotherms, which cannot maintain a constant internal temperature, and homeotherms, which maintain a nearly constant body temperature. Man is a homeotherm and endotherm: his thermoregulatory strategies are controlled by the hypothalamus, which acts as the body's thermostat. When it is hot, sweating, vasodilation of skin vessels and behavioural changes (light clothing, shade, rest) lose heat; when it is cold, shivering, vasoconstriction, and behavioural changes (warm clothing, shelter, activity) produce and conserve heat. Fever is a raised set point, and hyperthermia is an increased body temperature beyond the set point.

Unit 16: Support and Movement

Exercise - Section I: Objective Questions (MCQs)

Q1. Which one of the followings is not an unpaired bone: (Mandible and vomer are unpaired facial bones; sphenoid is an unpaired cranial bone; the nasal bones are paired.)

- Mandible
- Vomer
- Sphenoid
- ✓ Nasal

Q2. The fusion of four posterior vertebrae present in the pelvic region form:

- cervical
- lumber
- thorcic
- ✓ coccgial

Q3. Osteoarthritis is the most common chronic arthritis which is a degenerative joint disease also caused by: (Book text, p78: 'Sometimes, it may be caused by inheritance such as in osteoarthritis'.)

- ✓ Genetic defect
- Hormonal defect
- Nutritional cause
- Environmental cuase

Q4. Disease appearing due to low Ca+2 level in blood:

- Cramp
- Arthritis
- Spondylosis
- ✓ Tetany

Q5. The end of muscle which remains fixed when muscles contract:

- Insertion
- ✓ Origin
- Tendon
- Ligament

Q6. The human skeleton consists of:

- 80 bones
- 126 bones
- 200 bones
- ✓ 206 bones

Q7. The mature bone cells are called:

- Osteoblast
- ✓ Osteocytes
- Osteoclast
- Osteogenic

Q8. Cartilage is a connective tissue composed of: (NOTE: options (b) and (c) are both printed as '(b)' on the page; read as (b) Monocytes, (c) Myocytes.)

Lymphocytes

Monocytes

Myocytes

✓ Chondrocytes

Q9. Acetabulum provides articular surface for the:

✓ Femur

Humerus

Fibula

Radius

Q10. The number of bones in tarsal are:

8

5

✓ 7

3

Q11. Rheumatoid arthritis is inflammation of: (Book text, p78: 'Rheumatoid arthritis is the inflammation of hand and wrist joints'; finger joints are the hand-joint option.)

Knee bone

Elbow bone

✓ Finger bone

Big toe

Q12. Bone breaking cells are called:

Osteocytes

Osteoblast

Osteology

✓ Osteoclast

Q13. Human internal organs are protected mainly by:

✓ Axial skeleton

Appendicular skeleton

Hydro skeleton

Exoskeleton

Q14. Which of the following protein is the component of skeletal tissue:

Hemoglobin

Myoglobin

Fibrinogen

✓ Collagen

Q15. Stretching, tearing or twisting of ligaments is called:

Dislocation

✓ Sprain

Fracture

Reduction

Q16. The cardiac muscle cells are:

✓ Uninucleated

Polynucleated

Dinucleated

Non-nucleated

Q17. The end of muscle attached with moveable bone is called:

- Origin
- Flexing of muscle
- ✓ Insertion of muscle
- Belly of muscle

Q18. The diameter of actin protein filament is about: (Actin thin filaments are about 7 nm; 16 nm is the myosin thick filament.)

- 16nm
- 7mm
- 6cm
- ✓ 7nm

Q19. The in folding muscle fibre membrane is called:

- Sarcoplasmic reticulum
- Z-line
- Dark band
- ✓ T-tubules

Q20. Which ions are required for muscle contraction?

- Sodium
- Potassium
- ✓ Calcium
- Magnesium

Exercise - Section I: Complete the following sentences

Skeletal tissues are either bone or ____.

cartilage

Osteocytes are located in the small cavity called ____.

lacunae

The axial skeleton consisting of 80 bones and appendicular skeleton consists of ____ bones.

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The unique feature of hyoid bone is that it does not ____ with any other bone of the body.

articulate

The word clavicle is derived from Latin word ____.

clavicula

Ball and socket joint, hinge joint and pivot joint are the example of ____ joints.

synovial

Each intervertebral disc consists of an outer strong ring called annulus fibrous and soft gelatinous inner portion called ____.

nucleus pulposus

Arthritis is the inflammation of ____.

joints

When the joint is moved away from its original position, it is called joint ____.

dislocation

The study of muscle is called ____.

myology

The muscle is attached with bone by a connective tissue known as ____.

tendon

The sarcomere is the structural and functional unit of muscle ____.

contraction

The opposite movement of flexor and extensor muscles is called ____ movement.

antagonistic

Largest and complex joint of human body is ____ joint.

knee

The muscle that is contracting is called agonist and the muscle that is relaxing is called ____.

antagonist

Exercise - Section II: Short Questions

Q1. Describe the four types of cells associated with bone.

The four types of cells associated with bone are: (1) Osteogenic cells, the stem cells that divide to produce osteoblasts; (2) Osteoblasts, the bone-forming cells that secrete the bone matrix

and cause its mineralization; (3) Osteocytes, the mature bone cells that live in small cavities called lacunae between the lamellae and maintain the bone tissue; (4) Osteoclasts, the large bone-breaking (bone-resorbing) cells that break down bone tissue during remodelling and repair.

Q2. Differentiate between bone and cartilage.

Bone is a hard, rigid tissue impregnated with calcium salts; it is vascular (supplied with blood vessels and nerves) and provides strong support and protection, and its cells are osteocytes in lacunae within a matrix of collagen and minerals. Cartilage is a softer, flexible connective tissue composed of chondrocytes; it is not impregnated with calcium salts (except calcified cartilage), is generally avascular, and provides flexible support and smooth surfaces at joints, as in the nose, ear and ends of long bones.

Q3. Enlist the main functions of skeleton.

The main functions of the skeleton are: (1) Support — it gives the body its shape and framework; (2) Protection — it protects internal organs, e.g. the skull protects the brain and the rib cage protects the heart and lungs; (3) Movement — bones act as levers to which muscles attach for effective movement; (4) Production of blood cells — red bone marrow in spongy bone produces blood cells; (5) Storage of minerals — bones store calcium and phosphate salts; (6) Support for the body weight and maintenance of posture through the vertebral column.

Q4. Write the names of the cranium bones.

The cranium is made of 8 bones: frontal (1), parietal (2), temporal (2), occipital (1), sphenoid (1) and ethmoid (1). (Book Table 16.2.)

Q5. Write the names of facial bone.

There are 14 facial bones: 2 unpaired (mandible and vomer) and 6 paired (maxilla, zygomatic bones, nasal bones, inferior nasal conchae, lacrimal bones and palatine bones). They protect the sense organs of sight, smell and taste and provide a frame for muscle attachments.

Q6. Illustrate the three types of ribs.

The three types of ribs are: (1) True ribs (7 pairs, ribs 1-7), which attach directly to the sternum through their own costal cartilages; (2) False ribs (3 pairs, ribs 8-10), which attach to the costal cartilage of the rib above instead of directly to the sternum; (3) Floating ribs (2 pairs, ribs 11-12), which have no anterior attachment at all.

Q7. Differentiate between carpals and tarsals.

Carpals are the 8 small bones of the wrist, arranged in two rows, which give the wrist flexibility. Tarsals are the 7 larger, stronger bones of the ankle (including the calcaneus or heel bone and the talus), which bear the weight of the body and give the foot strength and stability. Thus carpals are smaller and more numerous for wrist movement, while tarsals are fewer, larger and built for weight-bearing.

Q8. Differentiate between Fibrous and cartilaginous joints.

Fibrous joints are those in which the articular surfaces of bones are connected by dense connective tissue consisting mainly of collagen; they are immovable joints because they allow no movement, e.g. the sutures between skull bones, the joint between a tooth and its socket, and the joint between the tibia and fibula. Cartilaginous joints are those in which the bones are connected by cartilage (fibrocartilage or hyaline cartilage); they are slightly movable joints because they allow a little movement, e.g. hyaline cartilage in the costal cartilages attaching ribs to the sternum, and fibrocartilage in the intervertebral discs and pubic symphysis.

Q9. Differentiate between two types of reductions.

The two types of reduction are: (1) Closed reduction, a procedure in which the bone fragments are coaxed back to their initial position by the physician's hands without surgical exposure of the bone fragments. (2) Open reduction, a procedure in which surgery is performed by dissecting the tissues and the fracture fragments are secured together with the help of pins or wires.

Q10. Write difference between simple, compound and complicated fractures.

(1) Simple (closed) fracture: the broken bones remain within the body, the skin remains intact and there is no damage to the surrounding tissues. (2) Compound (open) fracture: the broken bones penetrate the skin, or the force of injury breaks the skin, causing a risk of infection. (3) Complicated fracture: the broken bone damages the surrounding structures (organs), and there may be damage to arteries, veins and nerves; it heals more slowly.

Q11. Illustrate the first aid treatment for fractures.

First aid for a fracture: do not move the injured part unnecessarily; immobilize it with a splint and support it in a comfortable position; apply cold to reduce swelling and pain; control any bleeding with a clean cloth and do not push back any bone that pierces the skin; keep the patient calm and still; and get medical help at once so the fracture can be treated by reduction and immobilization.

Q12. Briefly explain sliding filament theory.

The sliding filament theory explains muscle contraction: a muscle fibre is made of myofibrils divided into sarcomeres, which contain thin actin filaments (about 7 nm) and thick myosin filaments (about 16 nm). When a nerve impulse arrives, calcium ions are released from the sarcoplasmic reticulum; they expose the binding sites on actin, the myosin heads attach to actin and, using energy from ATP, pull the thin filaments toward the centre of the sarcomere. The sarcomeres shorten, so the whole muscle fibre — and the muscle — contracts, while the filaments themselves do not shorten.

Q13. Define antagonistic arrangement of muscle.

Antagonistic arrangement means that muscles work in opposing pairs around a joint: when one muscle (the agonist) contracts to produce a movement, the opposite muscle (the antagonist) relaxes to allow it, and for the reverse movement their roles swap. For example, at the elbow the biceps contracts to flex the arm while the triceps relaxes, and the triceps contracts to extend the arm while the biceps relaxes. This paired action gives smooth,

controlled movement in both directions.

Q14. Write short note on muscle fatigue.

Muscle fatigue is the reduced ability of a muscle to contract after prolonged or repeated activity. It occurs because energy stores (ATP and glycogen) are used up, lactic acid and other metabolites accumulate in the muscle, and the nerve-muscle system becomes less responsive. The muscle feels weak and painful and cannot maintain force; rest, with removal of lactic acid and restoration of energy stores, recovers it.

Q15. Differentiate between tetany and cramp.

Tetany is a disease appearing due to a low level of Ca+2 in the blood: the nerves become hyperexcitable and the muscles undergo sustained, involuntary contractions (spasms) that do not relax normally. A cramp is a sudden, painful, involuntary contraction of a muscle or part of a muscle, usually brief, caused by overuse, dehydration or temporary electrolyte disturbance; unlike tetany it is not caused by a persistent low blood calcium level and it relaxes after a short time, often helped by stretching and rest.

Q16. Describe the sciatica.

Sciatica refers to back pain caused by a problem with the sciatic nerve, a large nerve that travels from the lower lumbar spine to the back of each leg. Any injury to the sciatic nerve causes pain in the lower back that travels to the hip, buttocks and leg. Common causes include a herniated (slip) disc, injury to the proximal sciatic nerve, spondylolisthesis (one vertebra slipping forward over

another), muscle spasm in the back or buttocks, and improper injection into the buttocks; pregnant women have a greater chance of a herniated disc and sciatica, and diabetes can also cause nerve damage. Recovery from sciatic injury is usually slow and incomplete.

Q17. Define arthritis and its types.

Arthritis is the inflammation of a joint: the joints become swollen, stiffer and painful, the membrane lining the joint thickens, and fluid production decreases, which increases friction. Its possible causes are infection or injury to the joints, abnormal metabolism and immune system dysfunction, and sometimes inheritance. Its types are: (1) Osteoarthritis, the most common type, in which the joint cartilage is degenerated; (2) Rheumatoid arthritis, the inflammation of hand and wrist joints; (3) Gouty arthritis, which develops in people with a high level of uric acid in their blood, caused by deposition of needle-like crystals of uric acid in a joint.

Exercise - Section III: Extensive Questions

Q1. Describe the structure of bone.

Bone tissue is of two types: compact (hard and dense) and cancellous (spongy). The outer part of a bone is hard compact bone, also called cortical bone, a hard white tissue surrounding all the bones; its fundamental units are osteons or Haversian systems, each consisting of concentric layers called lamellae, with a central (Haversian) canal in the centre containing the blood and nerve supply, communicating with the perforating (Volkmann's) canal that transmits vessels from the periosteum (the dense vascular

connective tissue enveloping the bone) into the endosteum (the thin layer of cells lining the medullary cavity). The osteocytes lie in small cavities called lacunae between the lamellae, and canaliculi are microscopic channels that transport nutrients to the osteocytes and remove their wastes. The inner part is spongy bone, also called cancellous or trabecular bone, a porous, highly vascular bone mostly at the ends of long bones; its lacunae lie in a lattice-like network of matrix spikes called trabeculae, its osteocytes are irregularly placed within the trabeculae, and the spaces between the trabeculae make it lighter and less dense than compact bone and contain red bone marrow, where blood cells are formed.

Q2. Explain the bones of fore limb in detail.

There are 30 bones in each forelimb: humerus (1), radius (1), ulna (1), carpals (8), metacarpals (5) and phalanges (14). The pectoral girdle connects the forelimb to the trunk: the clavicle (collar bone) and the scapula (shoulder bone, a flat triangular bone from the Latin 'scapul' meaning shoulder blade), whose lateral depression, the glenoid cavity, receives the head of the humerus. The humerus is the long bone of the upper arm; its head fits into the glenoid cavity at the shoulder, and its lower end forms the elbow joint with the radius and ulna of the forearm. The wrist has 8 carpals in two rows, the palm has 5 metacarpals, and the fingers have 14 phalanges (three in each finger and two in the thumb).

Q3. Define joint and explain different type of joints.

A joint is the place where two or more bones meet and movement takes place. Its types are: (1) Fibrous joints, connected by dense

connective tissue mainly of collagen, immovable, e.g. skull sutures, tooth sockets and the tibia-fibula joint. (2) Cartilaginous joints, connected by cartilage, slightly movable, e.g. intervertebral discs and pubic symphysis (fibrocartilage) and costal cartilages (hyaline cartilage). (3) Synovial joints, the most mobile type, in which the articular surfaces are covered with hyaline cartilage, a joint cavity filled with synovial fluid (secreted by the synovial membrane) lies between them, and ligaments and a joint capsule hold the joint; synovial fluid reduces friction and lubricates the surfaces. Synovial joints include the hinge joint (elbow and knee), ball and socket joint (hip and shoulder), gliding joint (between vertebrae), ellipsoid or condyloid joint (skull and 1st vertebra), pivot joint (atlas and axis) and saddle joint (carpometacarpal joint of the thumb).

Q4. Explain the steps of healing process of a simple fracture.

Bone healing is a natural process that usually takes 8-12 weeks (longer for large weight-bearing bones, slower in elderly people due to poor blood circulation) and takes place in four phases: (1) Hematoma formation — the first stage after fracture: injured blood vessels bleed at the fracture site, the blood clots and forms a hematoma, bone cells deprived of nutrients begin to die, and the tissues swell and cause much pain. (2) Soft callus formation — within a few days, capillaries grow into the hematoma and phagocytic cells clear the debris; fibroblasts and osteoblasts enter the site, fibroblasts produce collagen fibres connecting the ends of the broken bone while osteoblasts start forming spongy bone, and some bone spicules appear; the soft callus lasts approximately 4-6 weeks. (3) Hard (bony) callus formation — osteoblasts and

osteoclasts multiply rapidly, move inward and convert the soft callus into bony callus; woven bone is transformed into lamellar bone, and a ring of new bone tissue forms around the fracture, so the mended bone is slightly thicker there. (4) Bone remodelling — the bony callus is remodelled by osteoblasts and osteoclasts, unwanted bits are broken down and reabsorbed, and the final mended bone is almost undetectable on X-ray; remodelling may continue for months or even years.

Q5. Define antagonistic movement and explain the movement of knee joint.

Antagonistic movement is the paired action of muscles in which, when one muscle contracts, the opposite muscle relaxes, and vice versa, so movement is smooth and controlled in both directions. The knee joint is the largest and most complex joint of the body, a hinge-type synovial joint. Its movement is produced by antagonistic muscles: the quadriceps (extensor) group on the front of the thigh contracts to extend (straighten) the knee while the hamstring (flexor) group at the back relaxes, and the hamstrings contract to flex (bend) the knee while the quadriceps relax. Ligaments, the joint capsule and the patella (knee cap) stabilize the joint during this movement.

Q6. Describe the structure of skeletal muscle.

Skeletal muscle is made of bundles (fascicles) of long, cylindrical, multinucleated muscle fibres (cells) that are striated and voluntary. Each fibre is covered by a membrane, the sarcolemma, whose infoldings form T-tubules, and contains the sarcoplasmic reticulum

that stores calcium ions. Inside, the fibre contains myofibrils divided into sarcomeres, the structural and functional units of contraction, bounded by Z-lines. Each sarcomere has thin actin filaments (about 7 nm in diameter) and thick myosin filaments (about 16 nm); the regular overlap of these filaments gives the striated appearance, with dark bands (where thick and thin filaments overlap) and light bands. On stimulation, calcium ions released from the sarcoplasmic reticulum allow myosin heads to bind actin and pull the thin filaments toward the centre (sliding filament mechanism), shortening the sarcomere and contracting the muscle. The muscle is attached to bone by tendons; its fixed end is the origin and its movable end is the insertion.

Unit 17: Nervous Coordination

Exercise - Section I: Objective Questions (MCQs)

Q1. Nissel's granules is the group of:

- Mesosomes
- Lysosomes
- ✓ Ribosomes
- Chromosomes

Q2. Main transmitter for synapses that lie outside CNS is:

- Serotonin
- ✓ Acetylcholine
- Dopamine
- Adrenalin

Q3. In human rely center is located in: (NOTE: printed 'rely center' is read as relay centre — the thalamus, which lies in the fore brain.)

- ✓ Fore brain
- Mid brain
- Hid brain
- Spinal cord

Q4. The part of brain which controls breathing, heart rate and swallowing is:

- Cerebellum
- ✓ Medulla

Thalamus

Cerebrum

Q5. Which of the following receptors produces sensation of touch and pressure? (Book Summary, p124: 'The receptors for touch and pressure are also called mechanoreceptors'.)

Chemoreceptor

✓ Mechanoreceptors

Nociceptors

Photoreceptors

Q6. Chemoreceptors detect the stimulus of:

Pressure

Temperature

Light

✓ Smell

Q7. The portion of neuron that conducts nerve impulse towards cyton (cell body) is:

Axon

✓ Dendrites

Myelin

Soma

Q8. Neurilemma is the membrane of: (The neurilemma is the outer Schwann-cell membrane covering the nerve fibre.)

Muscle cell

cyton of neuron

Axon

✓ Nerve fibre

Q9. The fibres at the end of axon are called: (The terminal branches of the axon; 'telodendria' is not among the printed options.)

Dendrites

Dendron

✓ Axon fibres

Cyton

Q10. Synapse is microscopic gap between two successive:

Neuroglia

✓ Neuron

Axons

Soma

Q11. Brain and spinal cord together known as:

PNS

ANS

SNS

✓ CNS

Q12. Second largest part of brain is:

Fore brain

Cerebrum

✓ Cerebellum

Medulla

Q13. The number of cranial and spinal nerves in human are: (12 pairs of cranial nerves and 31 pairs of spinal nerves.)

- 24 and 62
- ✓ 12 + 31
- 20 + 68
- 20 + 62

Q14. The volatile organic chemicals, once commonly referred to as 'glue sniffing' are: (Book Summary, p125: 'Inhalants are volatile organic chemicals, commonly called glue sniffing'.)

- Cannabis
- ✓ Inhalants
- Narcotics
- Caffeine

Exercise - Section I: Fill in the blanks

Neurotransmitter molecules bind to the receptors on the ____ membrane at synapse.

post-synaptic

A microscopic gap between two consecutive neurons is called ____.

synapse

The collection of axons and dendrites are called ____.

nerves

Muscles and glands are principal ____.

effectors

An infection of the central nervous system is called ____.

meningitis

The structures which respond by impulse coming from motor neurons are ____.

effectors

Nociceptors detect stimulus of ____.

pain

Neurons are structural and functional unit of ____.

nervous system

Nissl's granules are the group of ____ in the neuron cell body.

ribosomes

Serotonin and acetylcholine are ____.

neurotransmitters

Exercise - Section II: Short Questions

Q1. Define nervous coordination, receptors and effectors.

Nervous coordination is the coordination of body activities by the nervous system through electrochemical signals (nerve impulses): receptors detect stimuli, the central nervous system interprets them, and effectors carry out responses. Receptors are the parts of the body that receive stimuli from the internal or external environment (e.g. sense organs). Effectors are the structures that

respond to impulses from motor neurons, principally muscles (which contract) and glands (which secrete).

Q2. Describe briefly about axon.

An axon is the long, single fibre of a neuron that conducts nerve impulses away from the cell body toward the next neuron, muscle or gland. It is usually covered by a myelin sheath formed by Schwann cells, with the outer neurilemma membrane and gaps called nodes of Ranvier, which speed conduction. At its end the axon branches into fine fibres (telodendria) ending in synaptic knobs that pass the impulse across a synapse.

Q3. Differences between sensory neuron and motor neuron.

Sensory neurons carry nerve impulses from receptors toward the central nervous system; their cell bodies lie outside the CNS (e.g. in dorsal root ganglia) and their dendrites are connected to receptors. Motor neurons carry impulses from the CNS toward effectors (muscles and glands); their cell bodies lie inside the CNS and their axons end at effectors. Thus sensory neurons bring information in, while motor neurons carry orders out.

Q4. Difference between reflex action and reflex arc.

Reflex action is the response itself: an immediate, involuntary and automatic response to an external or internal stimulus (e.g. withdrawing the hand from a hot object). Reflex arc is the pathway along which the impulse travels during that action: receptor, sensory neuron, centre in the CNS (brain or spinal cord, often via an associated neuron), motor neuron and effector. In short, reflex

action is the event and reflex arc is the route.

Q5. Define synapse, presynaptic neuron and post synaptic neuron.

A synapse is a microscopic junction (gap) between two successive neurons, which transmits the impulse from one neuron to another neuron. The presynaptic neuron is the neuron before the synapse, whose axon endings release the neurotransmitter. The postsynaptic neuron is the neuron after the synapse, whose membrane carries receptors that receive the neurotransmitter and pass the impulse onward.

Q6. Define neurotransmitter, write name of any four neurotransmitters.

A neurotransmitter is a chemical messenger released by the presynaptic neuron into the synaptic gap, which binds to receptors on the postsynaptic membrane and transmits the nerve impulse across the synapse. Four examples are acetylcholine, serotonin, dopamine and adrenalin (epinephrine). After acting, the transmitter is broken down (e.g. acetylcholine by acetylcholinesterase) so the synapse resets.

Q7. Write note on mid brain.

The mid brain is the middle part of the brain, lying between the fore brain and the hind brain. It contains the tectum, with centres for visual and auditory reflexes, the tegmentum, and the cerebral peduncles that carry nerve tracts between the fore brain and the hind brain and spinal cord. It relays visual and auditory impulses and controls movements of the eyes and body posture in response

to sights and sounds.

Q8. Describe ventricles of brain.

The ventricles of the brain are a series of connected cavities inside the brain filled with cerebrospinal fluid (CSF). There are four: two lateral ventricles (one in each cerebral hemisphere), the third ventricle in the diencephalon, and the fourth ventricle in the hind brain, connected by the cerebral aqueduct. The CSF cushions the brain, nourishes it and removes wastes, and the ventricles are lined by ependymal cells.

Q9. Name the five fundamental parts of reflex arc.

The five fundamental parts of a reflex arc are: (1) Receptor, which receives the stimulus; (2) Sensory neuron, which carries the impulse to the CNS; (3) Centre (intermediate or associated neuron in the brain or spinal cord), which interprets the impulse; (4) Motor neuron, which carries the response impulse away from the CNS; (5) Effector (muscle or gland), which carries out the response.

Q10. Describe structure and function of spinal card.

The spinal cord is a long, cylindrical cord extending from the medulla of the brain down the vertebral column, protected by vertebrae and the meninges. In cross-section it has inner grey matter (H-shaped, containing cell bodies) and outer white matter (nerve tracts). Its functions are to conduct impulses between the brain and the body (ascending sensory and descending motor tracts) and to serve as the centre for reflex actions; 31 pairs of spinal nerves arise from it.

Q11. Write any three differences between sympathetic and parasympathetic nervous system.

Three differences are: (1) The sympathetic system prepares the body for fight or flight (increases heart rate, dilates pupils, dilates bronchi, inhibits digestion), while the parasympathetic system restores rest-and-digest conditions (slows heart rate, constricts pupils, stimulates digestion). (2) Sympathetic nerves arise from the thoracic and lumbar regions of the spinal cord, while parasympathetic nerves arise from the brain and sacral region. (3) In the sympathetic system the preganglionic fibres are short and postganglionic fibres long, while in the parasympathetic system the preganglionic fibres are long and postganglionic fibres short.

Q12. Illustrate cranial nerves.

Cranial nerves are the 12 pairs of nerves that arise directly from the brain and form part of the peripheral nervous system. They are: olfactory (smell), optic (sight), oculomotor, trochlear and abducens (eye movements), trigeminal (face sensation and chewing), facial (facial expression and taste), vestibulocochlear or auditory (hearing and balance), glossopharyngeal (tongue and throat), vagus (heart, lungs and gut), accessory (neck and shoulder muscles) and hypoglossal (tongue movements). They may be sensory, motor or mixed.

Q13. Write note on gustatory receptors.

Gustatory receptors are the taste receptors: chemoreceptors located in the taste buds on the tongue (and parts of the mouth

and throat). They are stimulated by chemicals dissolved in saliva and produce the sensations of taste — sweet, sour, salty and bitter (and umami). The impulse travels through the facial, glossopharyngeal and vagus nerves to the gustatory centre of the brain, where taste is perceived.

Q14. Write short note on heroin.

Heroin is a powerful opioid narcotic drug made from morphine (obtained from the opium poppy). It is a depressant that produces euphoria, drowsiness and pain relief, but it is highly addictive: regular use causes tolerance and dependence, and overdose depresses breathing and can be fatal. It is taken by injection, smoking or snorting and causes severe health and social harm.

Q15. Prove that MRI is a better diagnostic test than CT scan.

MRI is a better diagnostic test than CT scan for nervous disorders for these reasons: (1) MRI uses strong magnetic fields and radio waves, not ionizing X-rays, so it avoids radiation exposure. (2) MRI gives much clearer images of soft tissues — brain, spinal cord and nerves — than CT, which is better for bones. (3) MRI can produce detailed images in several planes, revealing small tumours, strokes and demyelination that CT may miss. CT remains useful for fractures, bleeding and emergencies because it is faster and cheaper, but for soft-tissue nervous disorders MRI is superior.

Exercise - Section III: Extensive Questions

Q1. Describe structure of a typical neuron.

A typical neuron has three basic components: cell body (cyton or

soma), dendrites and axon. The cell body contains the nucleus and cytoplasm with Nissl's granules (groups of ribosomes) and mitochondria; it carries out the metabolic activities of the cell. Dendrites are short, branched fibres that receive stimuli and conduct impulses toward the cell body. The axon is a single long fibre that conducts impulses away from the cell body; it is covered by the myelin sheath (formed by Schwann cells) with the outer neurilemma, interrupted by nodes of Ranvier, and ends in fine branches (telodendria) with synaptic knobs. Based on function there are three types of neurons: sensory, motor and associated (intermediate) neurons.

Q2. Define reflex arc and explain it with example.

A reflex arc is the neural pathway of a reflex action: receptor, sensory neuron, centre in the CNS, motor neuron and effector. Example: when a hand touches a hot object, heat receptors in the skin are stimulated; the sensory neuron carries the impulse to the spinal cord, where an associated neuron passes it to a motor neuron; the motor neuron carries the impulse to the arm muscles (effectors), which contract and withdraw the hand at once, before the brain has consciously perceived the pain. The action is immediate, involuntary and automatic, and it protects the body from harm.

Q3. Write detail note on resting membrane potential.

Resting membrane potential is the electrical potential difference across the neuron membrane at rest, about -70 mV (inside negative relative to outside). It is maintained because the membrane is more

permeable to potassium ions, which leak out, leaving negative ions inside; the sodium-potassium pump actively transports 3 Na+ ions out and 2 K+ ions in, and the large negatively charged proteins inside the cell cannot cross the membrane. When the neuron is stimulated, sodium channels open, Na+ rushes in, the membrane depolarizes and an action potential travels along the fibre; afterwards the pump restores the resting potential.

Q4. Define synapse and describe structure and write steps of mechanism of synaptic transmission.

A synapse is a microscopic junction between two successive neurons that transmits the impulse from one neuron to another. Its structure has three parts: the presynaptic knob (axon ending) containing synaptic vesicles filled with neurotransmitter, the synaptic cleft (a narrow gap), and the postsynaptic membrane carrying specific receptors. The steps of synaptic transmission are: (1) the nerve impulse reaches the presynaptic knob; (2) calcium ions enter the knob; (3) synaptic vesicles fuse with the membrane and release neurotransmitter into the cleft; (4) the transmitter binds to receptors on the postsynaptic membrane and starts a new impulse (or inhibits it); (5) the transmitter is removed, e.g. acetylcholine is broken down by acetylcholinesterase, so the synapse resets for the next impulse.

Q5. Write the name of largest part of the brain and explain its structure and function.

The largest part of the brain is the cerebrum (of the fore brain). Its structure: two cerebral hemispheres connected by the corpus

callosum, an outer grey layer (cerebral cortex) with many folds (gyri and sulci) and inner white matter; each hemisphere has four lobes — frontal, parietal, temporal and occipital. Its functions: it is the seat of intelligence, memory, willpower and consciousness; it receives and interprets sensory information (sight, hearing, touch, taste, smell) and controls voluntary movements, speech, learning and reasoning.

Q6. Describe peripheral nervous system.

The peripheral nervous system (PNS) includes all the nerves outside the brain and spinal cord: 12 pairs of cranial nerves and 31 pairs of spinal nerves. It has two divisions: the somatic nervous system, which carries sensory information from receptors to the CNS and motor orders from the CNS to the skeletal muscles (voluntary actions), and the autonomic nervous system, which controls involuntary organs (heart, gut, glands) through its sympathetic (fight-or-flight) and parasympathetic (rest-and-digest) divisions. The PNS thus connects the CNS to receptors and effectors throughout the body.

Q7. Write note on olfactory receptors and taste receptors.

Olfactory receptors are the smell receptors: chemoreceptors in the olfactory epithelium of the nasal cavity, stimulated by chemicals dissolved in the mucus (liquid), which send impulses through the olfactory nerve to the olfactory centre of the brain. Taste (gustatory) receptors are chemoreceptors in the taste buds of the tongue and mouth, stimulated by chemicals dissolved in saliva, producing sweet, sour, salty and bitter tastes, with impulses carried

by the facial, glossopharyngeal and vagus nerves to the brain. Both are chemoreceptors because they respond to chemical stimuli rather than light, pressure or sound.

Q8. Write a comprehensive note on nicotine and cannabis.

Nicotine is an alkaloid derived from tobacco. It is a stimulant and highly addictive: it increases heart rate and blood pressure, constricts blood vessels and stimulates the nervous system; smoking also damages the lungs and causes cancer and heart disease. Cannabis (marijuana, hashish) comes from the Cannabis plant and contains tetrahydrocannabinol (THC); it acts as a depressant and mild hallucinogen, distorting perception, impairing memory, concentration and coordination, and causing dependence with regular use. Both drugs harm the nervous system and general health, and their use is a major cause of addiction among young people.

Q9. Describe stroke and meningitis.

Stroke is a sudden loss of brain function caused by interruption of the blood supply to part of the brain — either by a clot blocking a blood vessel (ischaemic stroke) or by bleeding into the brain (haemorrhagic stroke). The affected brain cells are damaged, causing sudden weakness or paralysis (often on one side), loss of speech, vision problems and confusion; it is a medical emergency. Meningitis is the inflammation of the meninges, the membranes covering the brain and spinal cord, usually due to bacterial or viral infection. Its symptoms are high fever, severe headache, stiff neck, vomiting and sensitivity to light, and it needs urgent treatment with

antibiotics (for bacterial meningitis) and supportive care.

Q10. Write note on electroencephalogram.

An electroencephalogram (EEG) is a record of the electrical activity of the brain, obtained by placing electrodes on the scalp which detect the tiny voltage changes produced by brain neurons. The record shows rhythmic waves (alpha, beta, theta and delta waves) that change with sleep, wakefulness and disease. EEG is commonly used to diagnose nervous disorders such as epilepsy (abnormal spike waves), sleep disorders, brain tumours and brain injuries, and to confirm brain death. It is painless, non-invasive and, unlike CT, uses no radiation.

Unit 18: Chemical Coordination

Exercise - Section I: Objective Questions (MCQs)

Q1. Excess MSH is secreted in: (In Addison's disease, low cortisol removes feedback inhibition, so ACTH and MSH from the same precursor rise and darken the skin.)

- ✓ Addison's disease
- Parkinson's disease
- Grave's disease
- Alzheimer's disease

Q2. The smaller number of a cell in pancreas secrete: (NOTE: the question wording is garbled as printed; insulin is the only pancreatic hormone among the options — gibberellins, cytokinins and ethene are plant hormones.)

- ✓ Insulin
- Gibberellins
- Cytokinins
- Ethene

Q3. Between the seminiferous tubules are interstitial cells, which secrete:

- Corticosteroids
- ✓ Testosterone
- Aldosterone
- Estrogen

Q4. Hormones are secreted by:

- Exocrine glands
- ✓ Endocrine glands
- Only from liver
- Only from pancreas

Q5. Insulin hormone is chemically made of:

- Steroid
- ✓ Protein
- Carbohydrates
- Nucleic acid

Q6. Aldosterone hormone in nature is:

- Protein
- Polypeptide
- ✓ Steroid
- Amino acid

Q7. Oxytocin and ADH are released by: (They are made in the hypothalamus and released from the posterior pituitary.)

- ✓ Posterior pituitary
- Hypothalamus
- Anterior pituitary
- Parathyroid glands

Q8. Which disease is not related with thyroid:

- Exophthalmic goitre

Myxoedema

✓ Diabetes mellitus

Cretinism

Q9. Alpha cells of pancreas secrete a hormone which is known as:

Insulin

✓ Glucagon

Thyroxin

Thymosin

Q10. Reabsorption of sodium and removal of potassium is promoted by:

✓ Aldosterone

FSH

Corticosterone

ADH

Q11. Male sex hormone is:

✓ Androgen

Progesterone

Estrogen

Parathormone

Q12. Positive feedback causes:

Stability

✓ Instability

Digestion

Respiration

Q13. Maintenance of level of glucose in the blood is:

- ✓ Negative feedback
- Sympathetic system
- Positive feedback
- Reflex action

Exercise - Section I: Fill in the Blanks

Corticosterone is chemically ____ in nature.

steroid

Ductless glands are also known as ____.

endocrine glands

Median pituitary lobe secretes the hormone ____.

MSH

Ovulation is induced by ____.

LH

Graves's disease is an example of ____.

hyperthyroidism

Lack of insulin causes a disease named ____.

diabetes mellitus

Reabsorption of water from collecting duct of nephron is promoted by ____.

ADH

Estrogen and progesterone are female ____ hormones.

sex

Gastrin is a hormone secreted by ____.

stomach

Labour is an example of ____ feedback mechanism.

positive

Exercise - Section II: Short Questions

Q1. Define hormone, write difference between endocrine and exocrine glands.

A hormone is a chemical messenger secreted by an endocrine gland into the blood, which travels to a distant target organ and controls its activity. Endocrine glands are ductless: they pour their secretions directly into the blood (e.g. pituitary, thyroid, adrenal). Exocrine glands possess ducts and pour their secretions onto a body surface or into a cavity through ducts (e.g. salivary glands, sweat glands, gastric glands). Some glands, like the pancreas, are heterocrine, having both parts.

Q2. Write role of any three steroid hormones.

Three steroid hormones and their roles: (1) Aldosterone (a mineralocorticoid from the adrenal cortex) promotes reabsorption of sodium and removal of potassium by the kidney, maintaining salt and water balance. (2) Cortisol/corticosterone (glucocorticoids)

raise blood glucose, help the body resist stress and reduce inflammation. (3) Testosterone (an androgen from the testes) develops male secondary sexual characters and supports spermatogenesis.

Q3. Illustrate the functions of FSH and LH.

FSH (follicle stimulating hormone) stimulates the growth of ovarian follicles and estrogen secretion in females, and spermatogenesis in the seminiferous tubules in males. LH (luteinizing hormone) triggers ovulation and the formation of the corpus luteum (which secretes progesterone) in females, and stimulates the interstitial (Leydig) cells of the testes to secrete testosterone in males.

Q4. Define acromegaly, write its causes.

Acromegaly is a disorder caused by over-secretion of growth hormone (GH) from the anterior pituitary after the growth plates of long bones have closed (in adults). Its cause is usually a benign tumour (adenoma) of the pituitary. Because the long bones can no longer grow in length, the bones of the face, jaw, hands and feet enlarge abnormally, and internal organs may also enlarge.

Q5. Write short note on oxytocin hormone.

Oxytocin is a hormone made in the hypothalamus and released by the posterior pituitary. It stimulates strong contractions of the uterus during labour (childbirth) and the ejection (let-down) of milk from the mammary glands during breastfeeding. Its release during labour is an example of positive feedback: contractions stimulate more oxytocin release until delivery.

Q6. Write role of insulin and glucagon.

Insulin, secreted by the beta cells of the islets of Langerhans, lowers blood glucose: it promotes glucose uptake by body cells, glycogen formation in liver and muscles, and inhibits breakdown of glycogen and formation of glucose. Glucagon, secreted by the alpha cells, raises blood glucose: it promotes breakdown of glycogen (glycogenolysis) and formation of glucose from non-carbohydrates (gluconeogenesis) in the liver. Together, by negative feedback, they keep blood glucose at a normal level; lack of insulin causes diabetes mellitus.

Q7. Define feedback mechanism.

A feedback mechanism is a control system in which the output of a process feeds back to regulate its own input, keeping a variable near its set point. In negative feedback the response reverses the deviation (e.g. high blood glucose triggers insulin, which lowers it). In positive feedback the response amplifies the deviation until an end point is reached (e.g. labour contractions stimulate more oxytocin until delivery).

Q8. Write similarities between chemical and nervous coordination.

Chemical and nervous coordination are similar in that: (1) both are coordination systems that link receptors, control centres and effectors; (2) both use messengers — hormones in blood and neurotransmitters across synapses — that act on specific receptors of target cells; (3) both maintain homeostasis by keeping internal conditions stable; (4) both are regulated by feedback mechanisms;

and (5) both can act on the same organs (e.g. the heart is controlled by nerves and by adrenalin). They differ in speed and duration: nervous coordination is fast and short-lived, chemical coordination is slow and long-lasting.

Q9. Describe the hormones secreted by kidney.

The kidney secretes: (1) Erythropoietin, which stimulates the bone marrow to produce red blood cells when oxygen is low. (2) Renin, secreted by juxtaglomerular cells when blood pressure falls; it starts the renin-angiotensin system that produces angiotensin and aldosterone, raising blood pressure and sodium retention. (3) Calcitriol (active vitamin D), which increases calcium absorption from the gut and helps regulate blood calcium with parathormone.

Q10. Write a note on thymus.

The thymus is a lymphoid gland situated behind the sternum (breastbone), large in children and shrinking after puberty. It secretes the hormone thymosin and is the site where T-lymphocytes mature: immature lymphocytes from the bone marrow migrate to the thymus, mature into T-cells, and then provide cell-mediated immunity, defending the body against infections and abnormal cells.

Exercise - Section III: Extensive Questions

Q1. Describe chemical composition of hormones.

Hormones fall into three chemical classes. (1) Steroid hormones, derived from cholesterol (lipids): e.g. cortisol, corticosterone, aldosterone and the sex hormones (testosterone, estrogen,

progesterone); they are lipid-soluble and pass through cell membranes to act on receptors inside cells. (2) Protein and polypeptide hormones: e.g. insulin, growth hormone, prolactin, parathormone, ADH and oxytocin; they are water-soluble and act on receptors on the cell membrane. (3) Amino-acid derivatives (amines): e.g. thyroxine (from tyrosine, containing iodine), adrenalin and noradrenalin, and melatonin. This chemistry decides how a hormone travels in blood and how it acts on its target cell.

Q2. Define endocrine gland, describe hormones of hypothalamus.

An endocrine gland is a ductless gland that secretes hormones directly into the blood, which carries them to distant target organs. The hypothalamus is the link between the nervous and endocrine systems: it secretes releasing and inhibiting hormones that control the anterior pituitary — thyrotropin-releasing hormone (TRH), gonadotropin-releasing hormone (GnRH), corticotropin-releasing hormone (CRH), growth-hormone-releasing hormone (GHRH), somatostatin (which inhibits GH) and dopamine (which inhibits prolactin). It also synthesizes ADH (vasopressin) and oxytocin, which are stored in and released from the posterior pituitary.

Q3. Write note on any four hormones of anterior pituitary.

Four hormones of the anterior pituitary: (1) Growth hormone (GH or somatotropin) — stimulates growth of bones and tissues and protein synthesis; its deficiency in childhood causes dwarfism and its excess causes gigantism in children and acromegaly in adults. (2) Thyroid stimulating hormone (TSH) — stimulates the thyroid gland to secrete thyroxine. (3) Adrenocorticotropic hormone (ACTH)

— stimulates the adrenal cortex to secrete cortisol and other corticoids. (4) Follicle stimulating hormone (FSH) — stimulates gamete formation (follicle growth in females, spermatogenesis in males); with LH it controls the gonads. (Prolactin, which stimulates milk production, and LH are also anterior pituitary hormones.)

Q4. Write note on thyroid gland.

The thyroid gland is a butterfly-shaped endocrine gland in the front of the neck, with two lobes joined by an isthmus, lying on the trachea. It secretes thyroxine (T4) and triiodothyronine (T3), iodine-containing hormones that regulate the basal metabolic rate, growth and development, and the activity of the nervous system; it also secretes calcitonin, which lowers blood calcium. Disorders: hypothyroidism in children causes cretinism (stunted growth and mental retardation) and in adults myxoedema (swelling, sluggishness, weight gain); hyperthyroidism causes Graves' disease or exophthalmic goitre (enlarged thyroid, protruding eyes, increased metabolism). Iodine deficiency causes simple goitre.

Q5. Write about parathyroid glands.

The parathyroid glands are four small glands embedded in the back of the thyroid gland. They secrete parathormone (PTH), which raises blood calcium: it promotes resorption of calcium from bones, reabsorption of calcium by the kidneys, and activation of vitamin D, which increases calcium absorption from the gut. Its action opposes calcitonin. Deficiency of PTH lowers blood calcium and causes tetany (sustained painful muscle spasms); excess PTH weakens bones and causes kidney stones.

Q6. Describe a comprehensive note on pancreas.

The pancreas is a heterocrine gland: its exocrine part (acini) secretes pancreatic juice with digestive enzymes into the duodenum, while its endocrine part, the islets of Langerhans, secretes hormones into the blood. The islets contain beta cells secreting insulin, alpha cells secreting glucagon, and delta cells secreting somatostatin. Insulin lowers blood glucose and glucagon raises it, so together they maintain glucose by negative feedback. Lack of insulin (or insensitivity to it) causes diabetes mellitus, with high blood glucose, glucose in urine, excessive thirst and urination, and weakness.

Q7. Describe hormones of adrenal cortex and also disorders of cortical hormones.

The adrenal cortex (outer part of the adrenal gland) secretes three groups of steroid hormones: (1) Mineralocorticoids, chiefly aldosterone, which promote sodium reabsorption and potassium excretion by the kidney, maintaining salt and water balance and blood pressure. (2) Glucocorticoids, chiefly cortisol and corticosterone, which raise blood glucose, help the body resist stress and suppress inflammation and immunity. (3) Sex corticoids (small amounts of androgens). Disorders: deficiency of cortical hormones causes Addison's disease (weakness, low blood pressure, loss of appetite and darkening of skin due to excess ACTH/MSH); excess of glucocorticoids causes Cushing's syndrome (obesity of face and trunk, high blood pressure and high blood sugar).

Q8. Write note on female sex hormones.

The female sex hormones are estrogen and progesterone, both steroids secreted by the ovaries. Estrogen, secreted by the developing ovarian follicles under FSH, develops the female secondary sexual characters (breast development, widening of hips, female voice and hair pattern) and regulates the menstrual cycle and growth of the uterus lining. Progesterone, secreted by the corpus luteum under LH, prepares the uterus for implantation of the embryo and maintains pregnancy, and inhibits further ovulation. Together, under the control of FSH and LH, they regulate the menstrual cycle and reproduction.

Q9. Define testosterone? Describe its function and effects of its deficiency.

Testosterone is the principal male sex hormone (an androgen), a steroid secreted by the interstitial (Leydig) cells of the testes under the stimulation of LH. Its functions: it develops the male secondary sexual characters at puberty (beard and body hair, deep voice, increased muscle mass and bone strength), supports spermatogenesis in the seminiferous tubules, and maintains libido. Effects of its deficiency: delayed or incomplete puberty, reduced muscle mass and strength, low libido, erectile problems and infertility, and in severe cases feminine features.

Q10. Define feedback mechanism; explain negative feedback mechanism with an example.

A feedback mechanism is a control system in which the output feeds back to regulate the input, keeping a variable near its set

point. In negative feedback the response opposes the change: for example, blood glucose regulation — after a meal, blood glucose rises, the beta cells of the pancreas secrete insulin, which promotes glucose uptake by cells and glycogen storage in the liver, so blood glucose falls back to normal and insulin secretion stops; when glucose falls too low, glucagon is secreted and raises it again. Thus the deviation is corrected and stability (homeostasis) is maintained.

Unit 19: Behavior

Exercise - Section I: Objective Questions (MCQs)

Q1. The scientific study of everything animals do, is called: (NOTE: the study itself is ethology, which is not among the printed options; behaviour is the subject studied, so it is the intended answer.)

- Pharmacology
- ✓ Behaviour
- Reflexes
- Society

Q2. The biorhythm, due to internal stimuli is: (Book text, p153: 'Internal Rhythm is also called endogenous, which is controlled by internal biological clock'.)

- Exogenous
- Diurnal
- ✓ Endogenous
- Crepuscular

Q3. The movement of Euglena towards dim light is: (Movement of the whole animal toward a stimulus is positive taxis.)

- ✓ Positive Taxis
- Negative Taxis
- Positive Kinesis
- Negative Kinesis

Q4. The useful behaviour for animal with short life span is: (Book text, p154: innate behaviour 'is especially beneficial for those animals which have short life span'.)

✓ Instinct

Learning

Reflex action

Taxis

Q5. The strongest and oldest adult male gorilla is called: (Book text, p163: 'The oldest and strongest adult male called the Silverback, is usually the dominant one of the troop'.)

✓ Silverback

King

Golden back

Beta

Q6. To solve a mathematical problem is an example of:

Instinct behaviour

✓ Insight learning

Reflex action

Involuntary act

Q7. A hormone like chemical acting outside the body is called: (Book Extra Information, p163: 'Pheromones are chemical messengers like hormones but act outside the body of secreting organism'.)

✓ Pheromone

Corticoid

Protein

Fatty acid

Q8. Trial and error learning is also called: (In this book's terminology, Type I = classical conditioning and Type II = operant/trial-and-error conditioning.)

Conditional reflex type I

Classical conditioning

✓ Conditional reflex type II

Insight learning

Q9. Organisms living together in organized groups are said to live in:

Colony

✓ Society

Solitary

Dependant

Q10. The animals having short life cycle depend on:

Learned behaviour

✓ Innate behaviour

Social behaviour

Agonistic behaviour

Exercise - Section I: Fill in the blanks

A ____ is a form of energy that activates a specific type sensory receptor.

stimulus

The behavioural activities of living organisms which take place at regular intervals are called ____.

biological rhythms

Hibernation is an example of ____ rhythm. (Book text, p154: 'Annual Rhythm: ... e.g. hibernation'.)

annual

There are two primary types of innate behaviour that is orientation and ____. (Book text, p154: 'There are two primary types of innate behaviours, i.e. orientation and non-orientation'.)

non-orientation

The movement of plant towards or away from stimulus of light is called ____.

phototropism

The time dependant form of learning involves. Exposure to sign stimuli, most often early in development is known as ____.

imprinting

Kohler, German psychologist performed many experiment on ____.

chimpanzees

A group of individuals of same species which interact and influence each other behaviour is called ____.

society

Adolescent females are transferred to a different troop once they reach about ____ years of age. (Book text, p163: 'Adolescent females, are transferred to a different troop once they reach about 8 years of age'.)
8

Aplysia, is marine snail were given the common name of ____ by ancient Greeks.
sea hare

Exercise - Section II: Short Questions

Q1. Define behaviour, ethology and ethologist.

Behaviour is everything an animal does in response to stimuli — all its actions and responses. Ethology is the scientific study of animal behaviour. An ethologist is a scientist who studies animal behaviour in natural conditions.

Q2. Differentiate between internal and external rhythm.

Internal (endogenous) rhythms are controlled by an internal biological clock, e.g. body temperature regulation and the menstrual cycle; they continue even without external cues. External (exogenous) rhythms are controlled by synchronizing the internal cycle with the external environment, e.g. sleep, wakefulness and the day/night cycle. In short, internal rhythms arise from within the organism, while external rhythms are driven by environmental cycles.

Q3. Define diurnal and annual rhythm.

Diurnal rhythm is a circadian (24-hour) rhythm synchronized with the day/night cycle in which the animal is active during the day (diurnal animals) or at night (nocturnal animals) or at twilight (crepuscular animals). Annual rhythm is a rhythm of about 365 days (less than or about a year), e.g. hibernation, reproduction in most animals and migration of birds.

Q4. Write short note on taxes.

Taxis (plural taxes) is a directional movement in which the whole animal either moves toward (positive taxis) or away from (negative taxis) a stimulus. For example, Euglena moves toward dim light (positive phototaxis). Taxis is an orientation behaviour: the direction of movement is related to the direction of the stimulus, unlike kinesis, in which movement is random in rate or turning and not directed toward or away from the stimulus.

Q5. Write three differences about instinctive and learning behaviour.

Three differences: (1) Instinctive (innate) behaviour is inborn and genetically programmed, while learning behaviour is acquired through experience. (2) Instinctive behaviour is performed in a fairly complete form the first time it is exhibited and is the same in all members of a species, while learning behaviour improves or changes with practice and varies between individuals. (3) Instinctive behaviour is especially useful for animals with a short life span and weakly developed nervous system, while learning behaviour needs a longer life and a more developed nervous

system (e.g. mammals and birds).

Q6. Describe the dances of bees.

Honeybees communicate by dances. The round dance is performed when food is near the hive (within about 50-100 metres): the bee moves in circles, indicating food nearby but not its direction. The waggle dance is performed when food is farther away: the bee moves in a figure-of-eight, waggling its abdomen; the direction of the waggle run relative to vertical indicates the direction of food relative to the sun, and the duration of the waggle indicates the distance. Other bees read the dance and fly to the food source.

Q7. Describe construction of intricate web by spider.

Spiders construct intricate webs by instinct, without learning. The spider first lays a bridge thread across a gap, then builds a frame and radial threads from a central hub, and finally spins a sticky spiral to catch prey, with a non-sticky spiral for its own movement. The web is a precise trap: vibrations in the threads tell the spider where prey is caught, and the spider rushes to wrap and feed on it. The whole construction is a genetically programmed sequence, performed perfectly the first time.

Q8. Write note on habituation.

Habituation is the simplest form of learning: an animal learns to ignore a repeated, irrelevant stimulus that has neither reward nor punishment. For example, the marine snail Aplysia (sea hare) withdraws its gill when touched, but after repeated harmless touches it stops responding; similarly, birds stop alarming at a

repeatedly harmless sound. Habituation lets animals save energy by not responding to unimportant stimuli.

Q9. Illustrate altruism.

Altruism means acting in the best interest of others rather than in one's own self-interest (some people believe altruism constitutes the essence of morality). In social insects, e.g. honeybees (Apis), the female workers are sterile and never produce offspring; instead they spend their whole lives looking after their siblings (brothers/sisters), especially helping their sister, the queen bee, to reproduce, thus aiding the production of new workers, drones and queens.

Q10. Define territorial behaviour.

Territorial behaviour is the behaviour by which an animal or a group of animals protects its territory from intrusions of other animals, especially members of the same species. Territorial boundaries may be marked by sounds such as bird songs, or by scent such as pheromones secreted by the skin glands of many mammals. Examples: mountain gorillas stand upright and pound their chests to intimidate intruders, and gibbon monkeys defend their territories with vigorous visual and vocal displays, their songs being heard up to a kilometre away.

Exercise - Section III: Extensive Questions

Q1. Discuss in detail biological rhythms.

Biological rhythms are behavioural activities of living organisms that take place at regular intervals, controlled by an internal

biological clock even without environmental stimuli. They may be internal (endogenous, e.g. body temperature regulation, menstrual cycle) or external (exogenous, e.g. sleep, wakefulness, day/night cycle). By period they are: circadian rhythms, endogenously generated rhythms with a 24-hour cycle (body temperature, sleep, wakefulness), controlled by the suprachiasmatic nuclei (SCN) in the hypothalamus; diurnal rhythms, circadian rhythms synchronized with the day/night cycle (diurnal, nocturnal and crepuscular animals); ultradian rhythms, much shorter than circadian (e.g. feeding cycle); and infradian rhythms, longer than 24 hours — monthly rhythms (e.g. the menstrual cycle in human females) and annual rhythms of about 365 days (e.g. hibernation, reproduction in most animals, migration of birds).

Q2. Define innate behaviour. Describe any two types of innate behaviour.

Innate behaviour (inborn or instinct behaviour) is behaviour preprogrammed within the nervous system before birth, demonstrated in a somewhat complete form the first time it is exhibited, without learning or altering by experience. Its two primary types are orientation and non-orientation behaviours. (1) Orientation behaviours are coordinated movements in response to an external stimulus, of two main types: taxis in animals (directional movement of the whole animal toward or away from a stimulus, e.g. Euglena toward light) and tropism in plants (growth toward or away from a stimulus, e.g. phototropism). (2) Non-orientation behaviours are innate behaviours not directed by a stimulus direction, such as reflexes and instincts — e.g. the human

suckling reflex of a newborn baby turning to suckle when touched on the cheek, or a spider building its web. Innate behaviour is especially beneficial for animals with a short life span and weakly developed nervous system, and all plants exhibit innate behaviour.

Q3. Describe migration of salmon and construction of hanging nest by some birds.

Salmon migration: salmon are born in freshwater streams, migrate to the sea to feed and grow, and return years later to the same stream to spawn, navigating by smell, the Earth's magnetic field and other cues; this long, instinctive journey ensures reproduction in safe, oxygen-rich water. Hanging nests: some birds (e.g. weaver birds) build hanging nests by instinct — the male weaves grass strips into a retort-shaped nest hanging from a branch, which protects eggs and chicks from predators and rain; the female chooses a mate partly by the quality of his nest. Both are complex innate behaviours performed without learning.

Q4. Explain mating behaviour of stickleback fish.

The mating behaviour of the three-spined stickleback fish is a chain of fixed, instinctive actions released by sign stimuli. In spring the male develops a red belly and builds a nest of weeds glued together, then performs a zigzag dance to attract a female. The female, swollen with eggs, follows him to the nest and lays eggs; the male then enters, fertilizes them, and chases the female away. He guards the nest, fans the eggs with his fins to oxygenate them, and protects the fry after hatching. Each step is triggered by the previous one (red belly, swollen belly, zigzag dance), showing a

fixed action pattern of innate behaviour.

Q5. Define learning behaviour. Explain imprinting.

Learning behaviour is behaviour that changes or is acquired through experience. Imprinting is a time-dependent form of learning: exposure to sign stimuli, most often early in development (a critical period), fixes a lasting attachment. The classic example is newly hatched ducklings or goslings following the first moving object they see (normally their mother) and treating it as their parent; Lorenz showed goslings imprinted on him and followed him. Imprinting is rapid, occurs only in a short sensitive period, and is usually irreversible.

Q6. Describe trial and error and spatial learning.

Trial and error learning (also called conditional reflex type II or operant conditioning) occurs through experience: the animal tries various responses, and those followed by reward are repeated while those followed by punishment are dropped — e.g. a rat in a Skinner box (designed by B. F. Skinner, an American psychologist) learns to press a lever for food. Spatial learning is learning about the layout of the environment: animals form a mental (cognitive) map of their surroundings, e.g. rats learning a maze, or bees learning the positions of landmarks, so they can find food, shelter and home efficiently.

Q7. Explain insight learning and habituation.

Insight learning is the sudden understanding of a problem and its solution without trial and error: Kohler, a German psychologist,

showed chimpanzees solving problems by insight — e.g. Sultan the chimpanzee stacked boxes to reach bananas, or joined sticks to pull food within reach — after apparently thinking about the situation. Habituation is the simplest form of learning: an animal learns to ignore a repeated, irrelevant stimulus, e.g. Aplysia (sea hare) stops withdrawing its gill after repeated harmless touches. Insight needs intelligence and a developed brain; habituation needs only a simple nervous system.

Q8. Define social behaviour, explain agonistic behaviour with example.

Social behaviour is behaviour involving interaction between individuals of the same species living together in organized groups (societies), e.g. wolves in packs or bees in hives, where individuals cooperate, communicate and divide labour. Agonistic behaviour (from Greek agon, struggle) is social behaviour in competitive situations: it includes threats or actual combat that settle disputes between individuals, e.g. two stags fighting for mates or territory, or dogs snarling and posturing over food. It maintains social order, such as territoriality and dominance hierarchy, but uses much energy and can cause injury, so ritualized threats usually settle disputes without real fighting.

Q9. Describe dominance hierarchy and altruistic behaviour.

Dominance hierarchy is a social structure within a group in which certain individuals are dominant over others and can claim better access to resources such as mates, shelter, food and other desirable commodities. Example: the pecking order in chickens,

determined by pecking (biting or striking with the beak); the top bird pecks all others, the second pecks all but the top, and so on, fixing each bird's status and role in the flock. Altruistic behaviour is acting in the best interest of others rather than oneself: e.g. sterile worker honeybees spend their lives rearing their siblings and helping the queen reproduce instead of producing their own offspring, and in gorilla troops the silverback protects the group. Altruism is explained by kin selection: helping relatives passes on shared genes.

Unit 20: Reproduction

Exercise - Section I: Objective Questions (MCQs)

Q1. The first convoluted part of vas deferens is: (The epididymis is the coiled tube in which the vas deferens begins and sperms are stored.)

- Vas deference
- ✓ Epididymis
- Sertoli
- Placenta

Q2. Decrease of FSH and increase of estrogen, causes pituitary to secrete: (Rising estrogen exerts positive feedback, triggering the LH surge that causes ovulation.)

- Luteotropic hormone (LTH)
- ✓ Luteinizing hormone (LH)
- Vasopressin (ADH)
- Oxytocin

Q3. Uterus opens into the vagina through:

- ✓ Cervix
- External genitalia
- Fallopian tube
- Vulva

Q4. Pregnancy is maintained by the:

Luteotropic hormone (LTH)

Corticosteroids

✓ Progesterone

LH and FSH

Q5. Syphilis is a sexually transmitted disease which is caused by:

Neisseria gonorrhoeae

✓ Treponema pallidum

E. coli

Myobacterium ovium

Q6. Reproductive system is different from other systems of body due to the fact that: (NOTE: option (c) is also true — it becomes functional at puberty — but the book's distinguishing point is that, unlike other systems, it is non-functional at birth.)

Functional at birth

✓ Non-functional at birth

Functional at puberty

Become functional in fetus

Q7. Fertilization occurs in:

Ovary

✓ Fallopian tube

Uterus

Vagina

Q8. The sperms are produced in the: (NOTE: options (c) and (d) are both printed as '(c)' on the page; read as (c) Seminiferous tubules,

(d) Prostate gland.)

Epididymis

Vas deferens

✓ Seminiferous tubules

Prostate gland

Q9. The cervix is the opening of:

Ovary

Vagina

Fallopian tube

✓ Uterus

Q10. Sperms are stored in:

✓ Epididymis

Urethra

Prostate gland

Vas deferens

Q11. FSH is a hormone produced by:

✓ Pituitary gland

Adrenal gland

Ovary

Testes

Q12. Uterus is a pear-shaped elastic sac about: (Book text, p173: 'Uterus is a pear shape elastic sac about 7.5cm long'.)

4cm long

5cm long

✓ 7.5cm long

10.5cm long

Q13. The average menstrual cycle of an adult human female is about:

5 days

8 days

✓ 28 days

14 days

Q14. Oligospermia is a condition in male in which: (Oligo- means few.)

Abnormal sperms are produced

More sperms are produced

No sperms are produced

✓ Less sperms are produced

Q15. The number of AIDS patients in world are in 2019: (Book Summary, p181: 'There are about 38 million people living with HIV infection at the end of 2019'.)

More than 1 billion

✓ Over 38 million

Less than 3 million

Less than 2 million

Exercise - Section I: Fill in the blank

Most part of male reproductive system located ____ of body.

outside

The scrotum acts as climate control for the ____.

testes

Female reproductive system is more ____ than male. (Book text, p173: 'The female reproductive system is more complex than the male reproductive system'.)

complex

There are ____ potential follicle cells already present at birth in ovary. (Book text, p173: 'approximately 4,000,00 potential (follicles) cells are already present at birth' — i.e. 400,000.)

400,000

The egg is spherical in shape and about ____ µm in diameter. (Book text, p173: 'about 120µm in diameter'.)

120

Ovulation is the release of mature ____ from the surface of the ovary.

egg

Chorionic gonadotropin hormone in urine is checked for ____ test.

pregnancy

IVF is commonly used technique to treat ____.

infertility

The natural or spontaneous termination of pregnancy is called ____.
miscarriage

The wilful ending of pregnancy is called ____.
abortion

Exercise - Section II: Short Questions

Q1. Illustrate that reproductive system is different from all other system.

The reproductive system is different from all other systems because it is non-functional at birth and becomes functional only at puberty, whereas other systems (digestive, respiratory, circulatory, nervous) function from birth. It is also the only system that is different in males and females, and it is not essential for the survival of the individual, but for the survival of the species through reproduction.

Q2. Write the functions of male reproductive system.

The functions of the male reproductive system are: to produce sperms in the seminiferous tubules of the testes; to produce the male sex hormone testosterone from the interstitial (Leydig) cells; to store sperms in the epididymis and nourish them with fluids from the seminal vesicles, prostate gland and Cowper's glands; and to deliver sperms into the female reproductive tract through the penis during copulation.

Q3. Describe that female reproductive system is more complex than male reproductive system.

The female reproductive system is more complex than the male

because the male needs only to produce and deliver gametes, while the female system, besides producing gametes, also gestates, gives birth to and nourishes the newborn. It is also under the influence of the menstrual cycle, with cyclic hormonal changes (FSH, LH, estrogen, progesterone), ovulation, menstruation and pregnancy — processes with no counterpart in the male system.

Q4. Write the causes of male infertility.

Causes of male infertility include: low sperm count (oligospermia) or no sperms (azoospermia); abnormal sperms; blockage of the sperm ducts; failure of the testes to descend or to produce testosterone; infections (e.g. mumps affecting the testes); exposure to heat, radiation or chemicals; smoking and alcohol; and hormonal disorders of the pituitary or testes.

Q5. Write the causes of female infertility.

Causes of female infertility include: failure to ovulate (hormonal disorders of the pituitary or ovaries); blocked or damaged fallopian tubes (often from infections); disorders of the uterus; endometriosis; age-related decline in egg quality; and infections of the reproductive tract, including sexually transmitted diseases.

Q6. Write some measures to prevent AIDS.

Measures to prevent AIDS include: avoid unprotected sexual intercourse and use protection; avoid sharing needles, syringes or razors; ensure blood for transfusion is screened for HIV; use sterilized instruments for medical, dental and piercing procedures; and spread awareness that HIV spreads through blood and sexual

contact, not through casual contact, food or mosquito bites.

Q7. Differentiate miscarriage and abortion.

Miscarriage is the natural or spontaneous termination of pregnancy — the end of pregnancy of its own, without deliberate action, usually due to abnormality of the fetus or health problems of the mother. Abortion is the wilful (deliberate) ending of pregnancy by medical or surgical means. Thus miscarriage is unintentional and abortion is intentional; both end the pregnancy before the fetus can survive independently.

Q8. Write the main causes of miscarriage.

The main causes of miscarriage are: chromosomal or genetic abnormalities of the fetus; infections of the mother; hormonal problems (e.g. low progesterone); abnormalities of the uterus or cervix; chronic diseases of the mother such as diabetes or thyroid disorders; trauma or injury; and exposure to harmful drugs, radiation or smoking.

Q9. Briefly describe AIDS.

AIDS (Acquired Immune Deficiency Syndrome) is the final stage of infection by HIV (human immunodeficiency virus), which attacks the body's immune system, especially the helper T-cells, and destroys the body's ability to fight infections and cancers. It spreads through sexual contact, blood (transfusion, shared needles) and from mother to child. There is no complete cure; antiretroviral drugs control the virus. About 38 million people were living with HIV infection at the end of 2019.

Q10. Describe the role of hormone to control the male reproductive function.

The male reproductive function is controlled by hormones in a negative-feedback loop. The hypothalamus secretes GnRH, which stimulates the anterior pituitary to secrete FSH and LH. FSH acts on the Sertoli cells and is necessary for sperm production (spermatogenesis); LH stimulates the Leydig cells to produce testosterone, which is necessary to continue spermatogenesis and develops male characteristics (mass and strength, fat distribution, bone mass, sex drive). Sertoli cells also produce inhibin, which controls spermatogenesis at a normal rate, and testosterone feeds back negatively on the hypothalamus and pituitary.

Exercise - Section III: Extensive Questions

Q1. Describe the structure and function of human male reproductive system.

The human male reproductive system consists of the testes, ducts and accessory glands, mostly located outside the body. The testes (two, in the scrotum, which acts as climate control keeping them cooler than body temperature) contain seminiferous tubules where sperms are produced, and interstitial (Leydig) cells secreting testosterone; Sertoli cells nourish the developing sperms. The first convoluted part of the vas deferens is the epididymis, where sperms are stored and mature; the vas deferens carries sperms to the urethra. Accessory glands add fluids: seminal vesicles produce a sugar-rich fluid giving sperms energy, the prostate gland contributes fluid to nourish and protect sperms, and Cowper's

(bulbourethral) glands produce a clear, slippery fluid that lubricates and neutralizes acidity in the urethra. The penis delivers sperms into the female tract. The whole system is controlled by GnRH, FSH, LH and testosterone with negative feedback.

Q2. Explain structure of female reproductive system.

The female reproductive system consists of ovaries, ducts and external genitalia. The ovaries (a pair, oval, attached to the dorsal body wall just below the kidneys) produce eggs: approximately 400,000 potential follicle cells are already present at birth, only about 500 ever mature, and usually one egg is released every month, alternately from the two ovaries, from puberty to menopause. The egg is spherical, about 120 µm in diameter, with a large nucleus and haploid chromosomes. The oviduct (fallopian or uterine tube) is a narrow muscular tube leading from the ovary to the uterus; the egg is fertilized in the oviduct. The uterus is a pear-shaped elastic sac about 7.5 cm long, the site of fetal development, with three wall layers (outer perimetrium, middle thick muscular myometrium, inner spongy endometrium); it opens into the vagina through the cervix. The external genitalia (vulva) protect the opening of the vagina.

Q3. Describe the hormonal control of female reproductive system.

The female reproductive system is controlled by the hypothalamus, pituitary and ovaries. The hypothalamus secretes GnRH, stimulating the anterior pituitary to secrete FSH and LH. FSH stimulates follicle growth and estrogen secretion; as estrogen rises and FSH falls, the pituitary secretes a surge of LH, which induces

ovulation (release of the mature egg from the ovary surface). The remaining follicle becomes the corpus luteum, which secretes progesterone that maintains the uterus lining and pregnancy; if fertilization does not occur, the corpus luteum degenerates, progesterone falls, and menstruation (elimination of the thickened uterus lining with bleeding through the vagina) occurs. Rising and falling levels of these hormones also exert negative and positive feedback on the hypothalamus and pituitary.

Q4. Describe the menstrual cycle and also explain its phases.

The menstrual cycle is the monthly cycle of the female reproductive system, averaging about 28 days in an adult female, controlled by FSH, LH, estrogen and progesterone. Its phases are: (1) Menstrual phase (days 1-5): the thickened endometrium is shed with bleeding through the vagina, as estrogen and progesterone are low. (2) Follicular (proliferative) phase (days 5-14): FSH stimulates a follicle to grow and secrete estrogen, which rebuilds the endometrium. (3) Ovulation (about day 14): the LH surge releases the mature egg from the ovary surface. (4) Luteal (secretory) phase (days 14-28): the corpus luteum secretes progesterone, which thickens the endometrium for implantation; if pregnancy does not occur, the corpus luteum degenerates and the cycle restarts with menstruation.

Q5. Define infertility and explain in vitro fertilization procedure to treat infertility.

Infertility is defined as not being able to get pregnant after one year or longer of unprotected sexual intercourse between partners.

It may be due to male causes (low or abnormal sperms, duct blockage, hormonal disorders) or female causes (failure to ovulate, blocked fallopian tubes, uterus disorders, infections). In vitro fertilization (IVF) is a common technique to treat infertility: eggs are collected from the woman's ovaries and fertilized by sperms in a laboratory dish (in vitro); the resulting embryos are grown for a few days and then transferred into the woman's uterus, where implantation and pregnancy can occur. The presence of chorionic gonadotropin (hCG) hormone in urine is checked for pregnancy test after the procedure.

Unit 21: Development and Ageing

Exercise - Section I: Objective Questions (MCQs)

Q1. The hypoblast is mainly presumptive: (The hypoblast, the lower layer of the blastoderm, gives rise to the endoderm.)

- ✓ Endoderm
- Mesoderm
- Ectoderm
- Blastoderm

Q2. At the cephalic end of primitive streak, closely packed cells from local thickening are known as: (Printed 'Hansen's node'; the standard name is Hensen's node.)

- Primitive gut
- Primitive ridges
- Splanchnic mesoderm
- ✓ Hansen's node

Q3. The cavity formed between somatic and splanchnic mesoderm is:

- Archenteron
- Henson's node
- ✓ Coelom
- Neurocoel

Q4. The morula is still about the same size of: (Cleavage divisions

increase cell number without growth, so the morula is the same size as the zygote.)

- Sperm
- ✓ Zygote
- Mulberry
- Blastocyst

Q5. New mothers experience cramps in uterus while nursing because of: (Oxytocin, released during suckling, causes uterine contraction and milk let-down.)

- Prolactin
- ✓ Oxytocin
- Both a and b
- hCG

Q6. Cleavage begins from zygote and end into:

- Morula
- Gastrulla
- ✓ Blastula
- Neurula

Q7. During cleavage, mulberry like compact ball is:

- ✓ Morula
- Blastulla
- Gastrula
- Blastoderm

Q8. Blastomeres are formed during:

Gastrulation

Neurilation

✓ Cleavage

Growth

Q9. The umbilical cord in human contains:

One artery, two vein

✓ Two arteries one vein

Only one vein

Only two vein

Q10. Study of ageing is called: (Book Summary, p211: 'The study of ageing is called gerontology'.)

Teratology

Paleozoology

✓ Gerontology

Biotechnology

Q11. Neurula tube is formed from:

✓ Ectoderm

Endoderm

Mesoderm

Hypoblast

Q12. Average gestation period in human female is: (NOTE: the book's Summary line says '266 days after last menstrual period', but 280 days (40 weeks) is the standard average gestation period and the intended MCQ answer.)

✓ 280 days

250 days

320 days

350 days

Q13. The fluid which surrounds embryo is called:

✓ Amniotic fluid

Chorionic fluid

Yolk

Uterus fluid

Q14. Prolactin prepare the mammary glands for the production of:

Sweat

Sebum

✓ Milk

Mucus

Exercise - Section I: Fill in the blanks

____ begins at around 12 or 13 years.

Puberty

Blastomeres are formed during ____.

cleavage

Neural plate is formed from ____.

ectoderm

The negative physiological changes in our body are said to be ____.

ageing

Ectoderm, mesoderm and endoderm are three ____ layers.
germ

Rubella, commonly known as German ____. (Book Summary, p211: 'Rubella is commonly known as German measles'.)
measles

The cleavage, in which entire egg divides into daughters' cells is called ____.
holoblastic

The pigment free area that appears at the time of fertilization is known as grey ____. (Book Summary, p211: 'Grey crescent is the area, which is important for proper development'.)
crescent

The upper layer of cells of blastoderm is called ____.
epiblast

The lower layer of cells of blastoderm is called ____.
hypoblast

Exercise - Section II: Short Questions

Q1. Define morula, blastula and gastrula.

Morula is the mulberry-like compact ball of cells (blastomeres) formed by cleavage of the zygote, still about the same size as the

zygote. Blastula (blastocyst in mammals) is the hollow ball of cells formed after cleavage, with a fluid-filled cavity (blastocoel) and an inner cell mass. Gastrula is the stage formed by gastrulation, in which cells move inward to form the three germ layers (ectoderm, mesoderm, endoderm) and the primitive gut (archenteron).

Q2. Write signs/symptoms of ageing.

Signs/symptoms of ageing include: greying and thinning of hair, wrinkling and loss of elasticity of skin, weakening of eyesight and hearing, loss of muscle strength and bone density, reduced efficiency of heart, lungs and kidneys, slower healing of wounds, decline in memory and learning, and reduced fertility (in man fertility gradually declines with age, while in female it ends with menopause).

Q3. Differentiate between meroblastic and holoblastic cleavage.

Meroblastic cleavage is partial cleavage: only part of the egg divides (in eggs with much yolk, e.g. birds and reptiles, cleavage is restricted to the small disc of cytoplasm at the animal pole). Holoblastic cleavage is complete cleavage: the entire egg divides into daughter cells (in eggs with little or no yolk, e.g. mammals and sea urchins). Thus meroblastic cleavage is incomplete and yolk-dependent, while holoblastic cleavage divides the whole egg.

Q4. Write fate of three embryonic germ layers.

Fate of the three embryonic germ layers: (1) Ectoderm gives rise to the nervous system (brain, spinal cord, neural tube, neural plate), the epidermis of skin, hair, nails, and sense organs. (2) Mesoderm

gives rise to muscles, skeleton, circulatory system (heart, blood), kidneys, gonads, and the coelom. (3) Endoderm gives rise to the lining of the digestive tract and respiratory tract, and associated glands (liver, pancreas, thyroid).

Q5. Describe changes in fetus during third trimester.

Changes in the fetus during the third trimester (months 7-9): the fetus grows rapidly and gains most of its weight; the lungs mature and the fetus practices breathing movements; the brain develops rapidly and the nervous system coordinates movements; the fetus turns head-down in preparation for birth; fat is deposited under the skin; and by the end of the third trimester the fetus is fully formed and ready for birth. Pregnancy is established when the blastocyst is implanted into the endometrium, and human pregnancy is divided into three trimesters.

Q6. Differences between identical twins and non-identical twins.

Identical twins develop from a single fertilized egg (zygote) that splits into two embryos: they have identical genes, are always of the same sex, and look alike. Non-identical (fraternal) twins develop from two separate eggs fertilized by two separate sperms: they share only about half their genes like ordinary siblings, may be of the same or different sexes, and do not necessarily look alike.

Q7. Write note on umbilical cord.

The umbilical cord is the cord connecting the fetus to the placenta. It contains two arteries and one vein: the umbilical vein carries oxygenated blood and nutrients from the placenta to the fetus,

while the two umbilical arteries carry deoxygenated blood and wastes from the fetus to the placenta. It is the lifeline of the fetus, supplying food and oxygen and removing wastes throughout pregnancy, and it is cut at birth.

Q8. Describe briefly about Rubella.

Rubella, commonly known as German measles, is a mild viral infection causing a rash, mild fever and swollen glands. It is dangerous during pregnancy: if a woman is infected in early pregnancy, the virus can cross the placenta and cause congenital rubella syndrome in the baby (deafness, blindness, heart defects, mental retardation). Vaccination (MMR) prevents rubella.

Q9. Illustrate the postnatal development.

Postnatal development has five life stages: neonate (newborn, first month), infant (up to about 2 years), childhood (about 2-12 years), adolescence (about 12-18 years; puberty begins at around 12 or 13 years), and adulthood (18 years onward, ending with ageing and death). After birth, growth continues, the child learns to walk and talk, puberty brings sexual maturity, adulthood is the reproductive phase, and old age brings declining body functions.

Q10. Write changes that occur at cellular level during ageing.

Changes at cellular level during ageing include: cells divide fewer times and accumulate damage to DNA; mitochondria become less efficient, producing less energy; harmful substances and waste pigments (e.g. lipofuscin) accumulate in cells; cell membranes and enzymes work less efficiently; and the ability to repair damage and

replace worn-out cells declines. These negative physiological changes in the body are collectively called ageing.

Exercise - Section III: Extensive Questions

Q1. Explain gastrulation in human.

Gastrulation in humans begins after the blastocyst implants into the endometrium. The inner cell mass forms a two-layered blastoderm: the upper epiblast and the lower hypoblast. Cells of the epiblast migrate through the primitive streak (with Hensen's node at its cephalic end) and move inward, forming the three germ layers: ectoderm (outer), mesoderm (middle) and endoderm (inner, from the hypoblast). The cavity formed between the somatic and splanchnic layers of mesoderm is the coelom, and the primitive gut (archenteron) forms inside. The first major phase of organogenesis is neurulation: the neural plate, formed from ectoderm, folds to form the neural tube, which becomes the brain and spinal cord. Thus gastrulation converts the hollow blastula into a three-layered gastrula, the basis of all organs.

Q2. Describe human embryonic development.

Human embryonic development proceeds as follows: fertilization in the fallopian tube forms the zygote; cleavage (holoblastic, the entire egg dividing) produces blastomeres, then the morula (mulberry-like ball, same size as the zygote), then the blastula/blastocyst with a cavity. The blastocyst implants into the endometrium and pregnancy is established. Gastrulation forms the three germ layers, and neurulation (the first major phase of organogenesis) forms the neural tube from ectoderm. By the start

of the third month the embryo is called a fetus. Organ systems develop through the three trimesters; the grey crescent, the pigment-free area appearing at fertilization, is important for proper development. The average gestation period is about 280 days, after which birth occurs, followed by postnatal development (neonate, infant, childhood, adolescence, adulthood).

Q3. Write role of hormones in controlling birth.

Birth is controlled by hormones in a sequence: toward the end of pregnancy, the level of progesterone (which maintains pregnancy) falls and estrogen rises, making the uterus more sensitive. The fetus and placenta release signals that stimulate the mother's posterior pituitary to secrete oxytocin, which causes strong rhythmic contractions of the uterine (myometrium) muscles. Oxytocin also causes the release of prostaglandins, which further increase contractions, in a positive-feedback loop. After birth, oxytocin causes uterine cramps during nursing and milk let-down, while prolactin prepares the mammary glands for milk production; lactation is the secretion and yielding of milk by the mother after giving birth.

Q4. Define ageing and describe factors of ageing.

Ageing is the sum of negative physiological changes in the body with advancing age, studied by gerontology. Its factors include: (1) Genetic factors — genes set the maximum lifespan and the rate of ageing. (2) Cellular factors — accumulation of DNA damage, reduced cell division, declining mitochondrial efficiency and accumulation of waste in cells. (3) Hormonal factors — declining

levels of growth hormone, sex hormones and others (e.g. menopause in females, gradual fertility decline in males). (4) Environmental factors — diet, smoking, radiation, pollution and stress accelerate ageing. (5) Disease factors — chronic diseases and infections weaken body systems. Signs include wrinkled skin, grey hair, weak eyesight and hearing, and reduced organ efficiency.

Q5. Describe the experiments of Spemann on the development of neural tube.

Spemann's experiments on the development of the neural tube (organizer experiment): Spemann and Mangold transplanted the dorsal lip of the blastopore (the organizer region, corresponding to Hensen's node) from one amphibian embryo into another embryo. The transplanted tissue induced the host tissue to form a second neural tube and a second embryo, showing that the organizer directs neighbouring cells to become neural tissue. This proved that the development of the neural tube is not self-determined but is induced by signals from the organizer, establishing the concept of embryonic induction — one group of cells controlling the fate of others.

Unit 22: Inheritance

Exercise - Section I: Objective Questions (MCQs)

Q1. All the genes found in a breeding population.

- Genotype
- Genome
- Gene frequency
- ✓ Gene pool

Q2. Incomplete dominance was discovered by 4'o clock plant in 1899 by: (NOTE: the year is as printed; Correns reported incomplete dominance in Mirabilis jalapa, the 4 o'clock plant, in 1900.)

- De-veries
- Johannsen
- ✓ Carl Correns
- Tschermach

Q3. Bilirubin damages brain cells and turns the skin and white of eyes yellow, condition is known as:

- Hepatitis
- Leukemia
- Botulism
- ✓ Jaundice

Q4. Green Colour Blindness is called: (Protanopia is red colour blindness, tritanopia is blue colour blindness.)

✓ Deuteranopia

Protanopia

Tritanopia

None of these

Q5. What is the risk of a color-blind child in a family when father is color-blind but mother is normal? (FLAGGED: read 'mother is normal' as homozygous normal, not a carrier — the book distinguishes a carrier female from a normal female. The colour-blind father gives his Y chromosome to his sons, so sons cannot inherit his X-linked allele, and he gives his affected X only to daughters, who receive a normal X from their mother and are carriers, not affected; so no child is colour-blind. If the mother were a carrier, 50% of sons would be affected.)

✓ Zero %

25%

50%

100%

Q6. The law of independent assortment can be determined by:

Test cross

Backcross

✓ Dihybrid cross

None of these

Q7. Each character is controlled by a pair of:

Chromosome

✓ Allele

Proteins

Enzymes

Q8. An allele is another word for: (NOTE: options (c) and (d) are both printed as '(c)' on the page; read as (c) Phenotype, (d) Different forms of a gene.)

Gene

Genotype

Phenotype

✓ Different forms of a gene

Q9. Phenotype refers to:

Genetic makeup

✓ Physical appearance

Recessive alleles

Dominant allele

Q10. The idea that different pairs of alleles are passed to offspring independently is Mendel's principle of:

Unit inheritance

Segregation

✓ Independent assortment

Law of probability

Q11. What type of inheritance do two alleles have if their traits blend together?

✓ Incomplete dominance

Co-dominance

Over dominance

Complete dominance

Q12. An allele whose trait only shows up when no dominant allele is present.

Hidden allele

Dominant allele

✓ Recessive allele

Present allele

Q13. This type of inheritance pattern shows both traits in possible offspring.

Complete dominance

Incomplete dominance

✓ Co-dominance

Dominant pattern

Q14. Which of the following gives information about phenotype but not genotype? ('Tall pea plant' describes appearance only — the plant could be TT or Tt — while the other options state genotypes.)

X^H Y

X^h Y

✓ Tall pea plant

Female carrier to colorblindness

Q15. If two white sheep produce a black offspring, the parent's genotype must be: (White is dominant; a black lamb needs a recessive allele from each parent, so both white parents must carry

it.)

- ✓ Heterozygous white
- Homozygous white
- Homozygous black
- Heterozygous black

Q16. One of the blood group would not be possible for children of a type AB mother and a type A father? (The AB mother has no i allele, so no child can be ii, i.e. group O.)

- ✓ Blood group O
- Blood group A
- Blood group B
- Blood group AB

Q17. All chromosomes other than sex chromosomes are called:

- Polysome
- ✓ Autosomes
- Mesosome
- Acrosome

Q18. If a gene is found on X-chromosome only then it is said to be:

- ✓ X-Linked
- Y-Linked
- Sex linked traits
- XY linked

Q19. The pattern of sex determination found in Drosophila is: (Book text, p239: the XY-XX type 'is found in human, drosophila and many

other animals'.)

WZ-ZZ Type

✓ XY-XX

XO-XX

Diploid, haploid type

Q20. The phenomenon of sex linkage was discovered by: (Book text, p241: 'T.H Morgan (1910) for the first time discovered sex linkage in Drosophila'.)

Carl Correns

Nilsson Ehle

✓ T.H. Morgan

Calvin Bridge

Exercise - Section I: Fill in the blanks

Alternate form of gene is known as ____.

allele

____ refers to presence of more than two allele at a locus.

Multiple alleles

The term gene was coined by ____.

Johannsen

ABO blood group is an example of ____.

multiple alleles

Polygenic inheritance, also known as ____.

quantitative inheritance

The pigment ____ is responsible for dark coloration in the skin.
melanin

A gene is said to be ____ when its presence suppresses the expression of a gene which is present on another locus.
epistatic

The Labrador retriever is highly popular type of ____.
dog

Genes that are located on the same chromosome are called ____.
linked genes

Drosophila has ____ pair of chromosomes.
4

In grasshopper male possesses unpaired X ____. (Book text, p240: 'In bugs and grasshopper, male possesses unpaired X-chromosome'.)
chromosome

The allele for haemophilia 'C' is located on ____. (NOTE: in standard genetics haemophilia C — factor XI deficiency — is autosomal, but this book treats haemophilia as X-linked, and the intended answer is the X-chromosome.)
X-chromosome

There are three basic types of cone cells which receive red, geen and ____ light.
blue

Exercise - Section II: Short Questions

Q1. Write the limitations of law of independent assortment.

Limitations of the law of independent assortment: (1) It applies only to genes located on different chromosomes, or far apart on the same chromosome. (2) It fails for linked genes — genes on the same chromosome tend to be inherited together and do not assort independently, unless crossing over separates them. (3) It does not apply to sex-linked genes in the same simple way, since their inheritance follows the sex chromosomes. (4) It assumes no interaction between genes, but epistasis and polygenic inheritance modify its ratios.

Q2. Describe briefly incomplete dominance.

Incomplete dominance is the inheritance pattern in which neither allele is completely dominant, so the heterozygous offspring shows a trait intermediate (blended) between the two parents. Example: the 4 o'clock plant (Mirabilis jalapa) — a cross between red-flowered (RR) and white-flowered (rr) plants gives pink-flowered (Rr) offspring in the F1 generation, and the F2 ratio is 1 red : 2 pink : 1 white, the same for phenotype and genotype.

Q3. Write short note on crossing over.

Crossing over is the exchange of corresponding segments between

non-sister chromatids of homologous chromosomes during prophase I of meiosis. It takes place at points called chiasmata and produces new combinations of alleles (recombinants). Crossing over breaks gene linkage, increases genetic variation, and is the basis of genetic mapping: the farther apart two genes are on a chromosome, the more likely crossing over occurs between them.

Q4. Describe two types of blood antigen make 4 types of blood group.

Two types of blood antigens make four types of blood group: antigen A and antigen B, present on the surface of red blood cells, controlled by three alleles (IA, IB, i) — an example of multiple alleles. Group A has antigen A (genotypes IAIA or IAi), group B has antigen B (IBIB or IBi), group AB has both antigens A and B (IAIB, co-dominance), and group O has neither antigen (ii). The plasma contains the opposite antibodies (anti-B in group A, anti-A in group B, neither in AB, both in O), which decides transfusion compatibility.

Q5. Erythroblastosis foetalis is a haemolytic disease of new born babies. Explain how?

Erythroblastosis foetalis is a haemolytic disease of newborn babies caused by Rh incompatibility. If an Rh-negative mother carries an Rh-positive fetus, fetal Rh-positive blood cells entering her circulation stimulate her to produce anti-Rh antibodies. In a later pregnancy with an Rh-positive fetus, these antibodies cross the placenta and destroy the baby's red blood cells, causing severe anaemia and jaundice (bilirubin damages brain cells and turns the skin and whites of the eyes yellow); in serious cases the baby may

die. It is prevented by giving the mother anti-Rh (anti-D) immunoglobulin after delivery.

Q6. Define epistasis and its types.

Epistasis is the phenomenon in which a gene at one locus suppresses (masks) the expression of a gene at another locus; the suppressing gene is called epistatic and the suppressed gene hypostatic. Its types are: (1) Dominant epistasis — a dominant allele masks another gene (e.g. 12:3:1 ratio). (2) Recessive epistasis — a recessive genotype (e.g. ee) masks another gene, as in Labrador coat colour, where ee dogs are yellow whatever their B/b genotype (9:3:4 ratio).

Q7. Differentiate between epistasis and dominance.

Dominance is interaction between alleles of the SAME gene at the same locus: one allele masks the other (e.g. T masks t in pea height). Epistasis is interaction between genes at DIFFERENT loci: a gene at one locus masks the expression of a gene at another locus (e.g. the E locus masks the B locus in Labrador coat colour). In dominance the masked allele is still present and can reappear in later generations; in epistasis the whole expression of the other gene is suppressed by the epistatic gene.

Q8. Describe Y linked inheritance in man.

Y-linked inheritance (holandric inheritance) is the inheritance of genes located on the non-homologous portion of the Y chromosome. Since only males have a Y chromosome, Y-linked genes are transmitted only from father to son, and every son of an

affected father is affected. Examples: hairy pinna (hypertrichosis of the ear) and genes for male fertility. Y-linked traits never appear in females and never skip a generation in the male line.

Q9. In human male determines the sex of offspring. Explain how?

In humans the male determines the sex of the offspring. The female produces only one type of egg (X), while the male produces two types of sperms (X and Y). If an X-bearing sperm fuses with the egg, the baby is a girl (XX); if a Y-bearing sperm fuses with the egg, the baby is a boy (XY). The ratio of male and female offspring is 1:1. In humans the male sex is determined by the SRY gene (sex determining region of Y chromosome) located on the short arm of the Y chromosome; absence of this gene in the zygote develops a female.

Q10. Describe sex limited traits.

Sex-limited traits are traits expressed in only one sex, although the genes for them are present in both sexes (e.g. milk production in female mammals, beard growth in human males, egg-laying in hens). They are autosomal traits whose expression is limited by sex hormones or sex-specific anatomy, unlike sex-linked traits, which are carried on the sex chromosomes. Sex-limited traits differ from sex-influenced traits, in which the same genotype is expressed differently in the two sexes (e.g. baldness).

Q11. Define monochromacy.

Monochromacy is total colour blindness: the person cannot distinguish any colours and sees only in shades of grey (black and

white). It occurs when two or all three types of cone cells (red, green, blue) are absent or non-functional, unlike dichromacy (e.g. protanopia, deuteranopia, tritanopia), in which only one cone type is defective.

Q12. Differentiate between X-linked recessive and X-linked dominant inheritance.

X-linked recessive inheritance: the recessive allele on the X chromosome is expressed in males with a single copy, but in females only when both X chromosomes carry it; females with one copy are carriers. It is more common in males than females, and a father cannot pass it to his son (pattern: grandfather to daughter, then grandson). Examples: haemophilia A and B, colour blindness.

X-linked dominant inheritance: a dominant allele on the X chromosome affects both sexes equally, but all daughters of an affected father are affected while no sons are; an affected mother passes it to 50% (one X) or 100% (both X) of children. Examples: hypophosphatemia (rickets), incontinentia pigmenti.

Q13. Write reason that X-linked recessive traits are more common in male than female.

X-linked recessive traits are more common in males than females because the male has only one X chromosome: if a recessive allele is present on that X, it expresses itself, there being no second X to mask it. A female possesses two X chromosomes, so she is affected only if she carries the allele on both; with one copy she is merely a carrier. Hence colour blindness and haemophilia appear far more often in males.

Q14. Describe XX-XY type pattern of sex determination.

The XX-XY type pattern of sex determination (male heterogamy): the male possesses a single X chromosome with a Y chromosome, while the female possesses two X chromosomes. The male produces two types of sperms (X and Y), the female only X eggs; an X sperm gives a girl (XX) and a Y sperm gives a boy (XY), in a 1:1 ratio. It is found in humans, Drosophila and many other animals, and in both groups the male determines the sex of the offspring.

Q15. Illustrate is WZ-ZZ type pattern of sex determination.

The WZ-ZZ type pattern of sex determination (female heterogamy): the female produces two types of gametes (W and Z), while the male produces only one type (Z). A Z sperm with a Z egg gives ZZ, which develops into a male; a Z sperm with a W egg gives WZ, which develops into a female, with equal chances of male and female offspring. In this pattern the female determines the sex of the offspring. It is found in birds and butterflies.

Q16. Compare sex determination in human and Drosophila.

Comparison of sex determination in human and Drosophila: in both there is male heterogamy (XY male, XX female) and the male determines the sex of the offspring. Differences: in humans sex is determined by the SRY gene on the short arm of the Y chromosome — absence of the Y gives a female, and an XXY human is a sterile male (Klinefelter's syndrome) while an XO human is a sterile female (Turner's syndrome). In Drosophila sex depends on the ratio of X chromosomes to autosomes, not on the Y: an XO Drosophila is a

sterile male and an XXY Drosophila is a fertile female. Thus sex chromosomes have less influence in Drosophila than in humans.

Q17. Differentiate between monohybrid and dihybrid cross.

A monohybrid cross is a cross between parents differing in one pair of contrasting characters (e.g. TT x tt for pea height), giving a 3:1 phenotypic ratio in F2. A dihybrid cross is a cross between parents differing in two pairs of contrasting characters (e.g. seed shape and colour), giving a 9:3:3:1 phenotypic ratio in F2 and demonstrating the law of independent assortment. A monohybrid cross tests segregation; a dihybrid cross tests independent assortment.

Q18. Differentiate between homozygous and heterozygous.

Homozygous means an organism carries two identical alleles for a character (e.g. TT or tt); it is pure and produces only one type of gamete for that character. Heterozygous means an organism carries two different alleles for a character (e.g. Tt); it is hybrid and produces two types of gametes. A homozygous dominant and a homozygous recessive look different, while a heterozygous organism shows the dominant trait but can pass the recessive allele to offspring.

Q19. Write reason that Y-linked inheritance is present only in male.

Y-linked inheritance is present only in males because the genes are located on the non-homologous portion of the Y chromosome, and only males possess a Y chromosome. A father transmits his Y chromosome to all his sons and to none of his daughters, so Y-linked traits (e.g. hairy pinna) pass directly from father to son and

never appear in females.

Q20. Write the reason that haemophilia and colorblindness cannot be transmitted from father to son directly.

Haemophilia and colour blindness cannot be transmitted from father to son directly because they are X-linked recessive traits: a father gives his Y chromosome (not his X) to his son, so the son cannot receive his father's X-linked allele. The son receives his X chromosome from his mother. The pattern is therefore grandfather to daughter (who becomes a carrier) and then to grandson, i.e. criss-cross inheritance.

Exercise - Section III: Extensive Questions

Q1. Define and explain Mendel's law of segregation.

Mendel's law of segregation states that each character is controlled by a pair of alleles (factors) which separate (segregate) during gamete formation, so that each gamete receives only one allele of the pair. In a monohybrid cross (TT x tt), the F1 plants are all heterozygous (Tt) and tall; on self-crossing, the alleles segregate again, giving F2 genotypes TT, Tt and tt in the ratio 1:2:1 and phenotypes tall and dwarf in the ratio 3:1. Segregation occurs because homologous chromosomes separate during meiosis, and it explains why recessive traits hidden in F1 reappear in F2.

Q2. Explain Mendel's Law of Independent assortment.

Mendel's law of independent assortment states that when two or more pairs of contrasting characters are considered, the alleles of different pairs assort independently of one another during gamete

formation. In a dihybrid cross (e.g. round yellow seeds RRYY x wrinkled green rryy), the F1 plants (RrYy) produce four types of gametes (RY, Ry, rY, ry) in equal numbers, giving an F2 phenotypic ratio of 9:3:3:1 (9 round yellow, 3 round green, 3 wrinkled yellow, 1 wrinkled green). The law is determined by a dihybrid cross and applies to genes on different chromosomes; it fails for linked genes on the same chromosome.

Q3. Explain in detail the ABO blood groups.

The ABO blood groups are controlled by three alleles at one locus (IA, IB, i) — an example of multiple alleles — with IA and IB co-dominant and both dominant over i. Four phenotypes result: group A (antigen A; genotypes IAIA or IAi), group B (antigen B; IBIB or IBi), group AB (both antigens A and B; IAIB, showing co-dominance, since both traits appear in the offspring), and group O (neither antigen; ii). The plasma carries the opposite antibodies: anti-B in A, anti-A in B, none in AB, both in O. Hence group O (no antigens) is the universal donor and group AB (no antibodies) the universal recipient. Rh factor (positive/negative) is inherited separately.

Q4. Define and explain multiple alleles.

Multiple alleles means the presence of more than two alleles of a gene at a locus in a population, although any one individual carries at most two. Example: the ABO blood group system, with three alleles IA, IB and i, giving six genotypes and four phenotypes (A, B, AB, O). Another example is coat colour in rabbits (four alleles). Multiple alleles arise by mutation of the same gene in different

ways and show dominance relationships: IA and IB are co-dominant with each other and dominant over i.

Q5. Write a note on transfusion principle of blood.

The transfusion principle of blood is based on antigens on the donor's red blood cells and antibodies in the recipient's plasma: a transfusion is safe only if the recipient's antibodies do not attack the donor's antigens. Group A (antigen A) can receive from A and O; group B from B and O; group AB (no antibodies) from all groups (universal recipient); group O (no antigens) can give to all groups (universal donor) but receive only from O. Cross-matching (testing donor cells against recipient serum) is done before transfusion, and Rh compatibility is also checked, since Rh-negative recipients must not receive Rh-positive blood.

Q6. Define erythroblastosis foetalis and describe its problem complications, causes and risk factors.

Erythroblastosis foetalis is a haemolytic disease of newborn babies caused by Rh incompatibility between an Rh-negative mother and an Rh-positive fetus. Problems and complications: the mother's anti-Rh antibodies cross the placenta and destroy the baby's red blood cells, causing severe anaemia, jaundice (bilirubin damages brain cells and turns the skin and whites of the eyes yellow), enlargement of liver and spleen, and in serious cases stillbirth or death of the newborn. Causes: sensitization of the Rh-negative mother by Rh-positive fetal blood, usually at a previous delivery, followed by antibody attack in a later Rh-positive pregnancy. Risk factors: Rh-negative mother with Rh-positive husband, previous

Rh-positive pregnancies, abortions or transfusions with Rh-positive blood. Prevention: anti-Rh (anti-D) immunoglobulin given to the mother after delivery.

Q7. Define polygenic inheritance and also explain wheat grain color as an example of polygenic inheritance.

Polygenic inheritance (quantitative inheritance) is inheritance in which a character is controlled by several genes at different loci, each contributing a small additive effect, so the trait shows continuous variation rather than distinct classes. Example — wheat grain colour: studied by Nilsson-Ehle, it is controlled by two genes (or three pairs); a cross between dark red and white grains gives F2 shades from dark red to white in a ratio depending on the number of contributing alleles, with intermediate shades most common. Human skin colour and height are similar examples: more dominant alleles give darker skin, and the trait varies continuously in the population.

Q8. Explain epistasis with reference to inheritance of coat color in Labrador retriever.

Epistasis in Labrador retriever coat colour: coat colour is controlled by two loci — B/b for pigment type (B black dominant, b chocolate) and E/e for pigment deposition. The E locus is epistatic: a dog with genotype ee cannot deposit pigment in the hair and is yellow, whatever its B/b genotype (BBee, Bbee and bbee are all yellow). Thus black dogs are B_E_, chocolate dogs are bbE_, and yellow dogs are __ee, giving a 9:3:4 ratio in a dihybrid cross. The Labrador retriever, a highly popular type of dog, is the classic example: the

recessive epistatic gene ee masks the colour gene B/b.

Q9. Describe Gene linkage by the help of Morgan's experiment.

Gene linkage (Morgan's experiment): genes located on the same chromosome are called linked genes and tend to be inherited together, violating independent assortment. T. H. Morgan (1910) crossed Drosophila for body colour and wing size and found that parental combinations appeared far more often than recombinants in the offspring, proving that the genes for these traits are linked on the same chromosome. Recombinant types appeared only when crossing over exchanged segments between homologous chromosomes, and the frequency of recombination showed how far apart the genes lie — the basis of chromosome mapping.

Q10. Define sex linkage and explain sex linkage in Drosophila.

Sex linkage is the phenomenon in which genes located on the sex chromosomes (X or Y) control traits not concerned with sexual characters, e.g. the gene for blood clotting factor VIII, the gene for opsin pigment in the eye, and the gene for hairy pinna. Sex linkage in Drosophila was discovered by T. H. Morgan (1910): a white-eyed male appeared in a red-eyed culture; crossed with a red-eyed female, all F1 flies were red-eyed, but in F2 the red and white-eyed flies appeared in a 3:1 ratio, with all white-eyed flies being male. Morgan concluded that the white-eye gene is recessive and located on the X chromosome: males, having one X, express it, while females need two copies.

Q11. Explain color blindness and haemophilia as examples of

X-linked recessive inheritance.

Colour blindness and haemophilia are examples of X-linked recessive inheritance. Colour blindness (defective opsin pigment in cone cells: protanopia red, deuteranopia green, tritanopia blue; monochromacy is total colour blindness) occurs when the recessive allele is present on the single X of a male or on both X chromosomes of a female; females with one copy are carriers. Haemophilia (A: factor VIII deficiency; B: factor IX deficiency) prevents normal blood clotting, so a small injury causes prolonged bleeding. Both traits are far more common in males, a father cannot pass them to his son (he gives his Y to sons), and they pass from grandfather through a carrier daughter to grandson.

Q12. Human skin color is an example of polygenic inheritance. Explain.

Human skin colour is an example of polygenic (quantitative) inheritance: it is controlled by several genes at different loci, each contributing melanin pigment in an additive way, so skin colour varies continuously from very light to very dark rather than in distinct classes. The pigment melanin is responsible for dark coloration of the skin; more dominant alleles at the contributing loci give more melanin and darker skin. Children of parents with different skin colours typically show intermediate shades, and the distribution of shades in the population follows a bell-shaped (normal) curve, characteristic of polygenic traits.

Unit 23: Chromosome and DNA

Exercise - Section I: Objective Questions (MCQs)

Q1. Morphological characteristics of chromosomes are collectively called:

- Holotype
- Karyokinesis
- ✓ Karyotype
- Neotype

Q2. Mongolism is also known as:

- ✓ Down's syndrome
- Klinefelter's syndrome
- Turner's syndrome
- Jacob's syndrome

Q3. All are related to turner's syndrome, except: (Turner's syndrome features are short stature, webbed neck and underdeveloped (streak) ovaries; a broad face is a feature of Down's syndrome, not Turner's.)

- Short stature
- Webbed neck
- ✓ Broad face
- Without ovaries

Q4. The human chromosomes are composed of:

DNA+RNA

DNA+Lipid

✓ DNA+RNA+Protein

DNA only

Q5. Synthesis of a new DNA strand begins with:

a DNA primer

✓ an RNA primer

an Okazaki fragment

DNA polymerase I

Q6. The chromosomes were first discovered in 1882, by: (NOTE: Waldeyer coined the name 'chromosome' in 1888; Fleming observed chromosomes and mitosis.)

Morgan

✓ Walther Fleming

Carl Correns

Waldeyer

Q7. The number of chromosomes in Monkey is: (NOTE: book-table value; chromosome numbers vary by monkey species.)

46

✓ 48

50

44

Q8. The strand of DNA that elongate towards the replication fork:

Lagging

✓ Leading

Primer

Okazaki

Q9. The number of tRNA in human are: (61 sense codons code amino acids, so 61 tRNA types are needed in theory.)

45

60

✓ 61

35

Q10. Anticodons are present on:

mRNA

✓ tRNA

DNA

rRNA

Q11. The initiation codon is:

UUA

ACC

✓ AUG

GCG

Q12. Down's syndrome is autosomal non-disjunction of chromosome number:

✓ 21

22

13

18

Q13. Which condition appears due to point mutation: (Turner's, Klinefelter's and Down's syndromes are chromosomal non-disjunction disorders, not point mutations.)

Turner's syndrome

Klinefelter's syndrome

✓ Sickle cell anaemia

Down's syndrome

Exercise - Section I: Fill in the blanks

The name chromosome was assigned by ____. (Waldeyer coined the term in 1888.)

Waldeyer

The number of chromosome in frog is ____. (Standard textbook value; numbers vary by frog species.)

26

The copying of mRNA from DNA is called ____.

transcription

The strand of DNA that runs from replication fork is called ____. (The leading strand elongates toward the fork; the lagging strand runs away from it.)

lagging strand

The sequence of 3 nucleotides on mRNA is called ____.

codon

Both UUU and UVC are codon of ____ amino acid. (NOTE: 'UVC' is a misprint for UUC; both UUU and UUC code phenylalanine.)
phenylalanine

The syndrome having trisomy of chromosome No.21 is ____ syndrome.
Down's

Phenylketonuria is an example of ____ mutation.
point

If a person has 44 autosomes + XXY, he will suffer from ____.
Klinefelter's syndrome

Change in DNA structure is called ____.
mutation

Exercise - Section II: Short Questions

Q1. Differentiate between heterochromatin and euchromatin.

Heterochromatin is the densely packed, darkly staining part of chromatin that is genetically inactive (genes switched off) and replicates late; it is found mainly near centromeres and telomeres.
Euchromatin is the loosely packed, lightly staining part of chromatin that is genetically active (genes switched on) and replicates early; most transcription takes place in euchromatin.
During interphase, heterochromatin stays condensed while

euchromatin is extended.

Q2. Write any three parallel behaviors between genes and chromosome.

Three parallel behaviours between genes and chromosomes: (1) Both genes and chromosomes occur in pairs in diploid cells and exist singly in gametes. (2) Both segregate during meiosis — members of a gene pair and homologous chromosomes separate into different gametes. (3) Both show independent assortment — different gene pairs and different chromosome pairs assort independently. These parallels led Sutton and Boveri to propose the chromosomal theory of inheritance.

Q3. Explain types of chromosomes on the basis of centromeric position.

Types of chromosomes on the basis of centromere position: (1) Metacentric — centromere in the middle, arms of equal length, V-shaped. (2) Submetacentric — centromere slightly off-centre, arms unequal, L-shaped. (3) Acrocentric — centromere near one end, one arm very short, J-shaped. (4) Telocentric — centromere at the very end, only one arm, rod-shaped. The centromere divides the chromosome into two arms (p short arm and q long arm) and attaches spindle fibres during division.

Q4. Differentiate between conservative model and dispersal model of DNA replication.

Conservative model of DNA replication: the original double helix stays intact and a completely new double helix is made — one

daughter molecule is entirely old, the other entirely new. Dispersal (dispersive) model: the parental DNA is broken into fragments and the daughter molecules contain a mixture of old and new segments scattered along both strands. Both models were disproved by the Meselson-Stahl experiment, which supported the semi-conservative model (each daughter helix has one old and one new strand).

Q5. Define RNA primer, primase and ligase.

RNA primer is a short stretch of RNA (about 5-10 nucleotides) synthesized by primase on the DNA template, providing the free 3'-OH end that DNA polymerase needs to begin synthesis of a new DNA strand. Primase is the enzyme (a type of RNA polymerase) that synthesizes the RNA primer. Ligase is the enzyme that joins DNA fragments — it seals the nicks between Okazaki fragments on the lagging strand, forming continuous phosphodiester bonds.

Q6. Write difference between leading strand and lagging strand.

Leading strand: the new DNA strand synthesized continuously in the 5' to 3' direction toward the replication fork, needing only one RNA primer. Lagging strand: the new DNA strand synthesized discontinuously in short pieces (Okazaki fragments) away from the replication fork, needing many RNA primers, which are later removed and the fragments joined by ligase. The leading strand elongates toward the fork; the lagging strand runs from the fork.

Q7. Write any three characteristics of codons.

Three characteristics of codons: (1) A codon is a sequence of three nucleotides (a triplet) on mRNA that codes for one amino acid (e.g.

AUG for methionine). (2) The genetic code is universal (almost the same in all organisms), specific (each codon codes a particular amino acid) and degenerate (more than one codon can code the same amino acid, e.g. UUU and UUC both code phenylalanine). (3) There are 64 codons: 61 sense codons coding amino acids and 3 stop codons (UAA, UAG, UGA) that terminate translation; AUG is the initiation codon.

Q8. Describe importance of regulation of gene expression.

Regulation of gene expression is important because: (1) It allows a cell to produce only the proteins it needs, when it needs them, saving energy and resources. (2) It enables cell differentiation — different cells express different genes (e.g. a muscle cell expresses muscle proteins, not haemoglobin). (3) It lets organisms respond to environmental changes (e.g. bacteria switching on the lac operon in the presence of lactose). (4) It controls growth and development, ensuring genes act in the right order at the right time.

Q9. Write differences between physical and chemical mutagens.

Physical mutagens are physical agents causing mutation, e.g. ultraviolet (UV) rays, X-rays and gamma rays (ionizing radiations) that break DNA strands or form abnormal bonds between bases. Chemical mutagens are chemical agents causing mutation, e.g. nitrous acid, mustard gas, formaldehyde, benzene and certain dyes, which alter base structure or cause wrong base pairing during replication. Physical mutagens act by radiation energy; chemical mutagens act by chemical reaction with DNA bases.

Q10. Describe useful aspects of mutation.

Useful aspects of mutation: (1) Mutations are the raw material of evolution — they create new alleles and genetic variation on which natural selection acts. (2) They are used in plant and animal breeding to produce improved varieties (e.g. disease-resistant crops, high-yield wheat). (3) They help in research and medicine, e.g. understanding gene function and producing useful mutants in microorganisms for antibiotics and industrial products.

Q11. Describe harmful aspects of mutation.

Harmful aspects of mutation: (1) Most mutations are harmful — they disrupt normal proteins and cause genetic diseases (e.g. sickle cell anaemia from a point mutation, phenylketonuria). (2) Chromosomal mutations cause syndromes (e.g. Down's syndrome from trisomy 21, Turner's and Klinefelter's syndromes). (3) Mutations in body cells can cause cancer by disturbing the control of cell division. (4) They can cause death of the embryo or severe disabilities.

Q12. Write cause of sickle cell anaemia.

Cause of sickle cell anaemia: a point mutation in the gene for the beta chain of haemoglobin, in which the codon GAG (glutamic acid) is changed to GUG (valine) at the sixth position of the beta chain. The abnormal haemoglobin (HbS) makes red blood cells sickle-shaped under low oxygen; the sickled cells block capillaries, break down easily and cause anaemia, pain and organ damage. It is an autosomal recessive disorder.

Q13. Write treatment of phenylketonuria is possible.

Treatment of phenylketonuria is possible: PKU is caused by the lack of the enzyme phenylalanine hydroxylase, so phenylalanine accumulates and damages the brain. If detected early (newborn screening), the child is kept on a strict low-phenylalanine diet (avoiding protein-rich foods and aspartame) from infancy, which prevents mental retardation and allows normal development. The diet must be followed especially in childhood, when the brain is developing.

Q14. Write the symptoms of Down's Syndrome.

Symptoms of Down's syndrome: flat (broad) face with slanting eyes and small ears, short stature, short neck, protruding tongue, single transverse palmar crease, poor muscle tone (hypotonia), mental retardation of varying degree, congenital heart defects, and increased susceptibility to infections and leukaemia. It is caused by trisomy of chromosome 21 (47 chromosomes) due to non-disjunction, and is also called Mongolism.

Q15. Write symptoms of Turner's syndrome.

Symptoms of Turner's syndrome: short stature, webbed neck, broad (shield) chest with widely spaced nipples, underdeveloped (streak) ovaries leading to infertility and absence of menstruation, and delayed or absent puberty. Other features may include swelling of hands and feet at birth, heart and kidney defects, and normal intelligence. It is caused by the absence of one X chromosome (45, XO) due to non-disjunction.

Exercise - Section III: Extensive Questions

Q1. Explain structure of chromosome.

A chromosome is a thread-like structure in the nucleus, made of chromatin (DNA + RNA + protein, mainly histone proteins), visible during cell division. Its structure: two sister chromatids joined at the centromere (primary constriction), which divides it into a short arm (p) and a long arm (q). The ends are called telomeres, which protect the chromosome. Some chromosomes have a secondary constriction (nucleolar organizer) and a satellite (SAT chromosome). On the basis of centromere position, chromosomes are metacentric (middle), submetacentric (slightly off-centre), acrocentric (near one end) and telocentric (at the end). Human cells have 46 chromosomes (44 autosomes + 2 sex chromosomes); morphological characteristics of chromosomes are collectively called the karyotype.

Q2. Describe in detail the composition of chromosome.

Composition of a chromosome: a human chromosome is composed of DNA, RNA and protein. DNA is the genetic material, forming a double helix of two polynucleotide strands with complementary base pairing (A-T, G-C). Histone proteins (H2A, H2B, H3, H4) form octamers around which DNA winds (about 200 base pairs per nucleosome) to form the beads-on-a-string structure, coiled further into the 30 nm fibre and the chromatid. Non-histone proteins (scaffold proteins, enzymes) maintain structure and control gene activity. A small amount of RNA is also present. Thus DNA provides the information, while proteins package it and regulate its expression.

Q3. Illustrate the experiment performed by Hershey-Chase.

Hershey-Chase experiment (1952): Hershey and Chase worked with bacteriophage T2, which infects bacteria. They grew one batch of phages in radioactive sulphur (35S, labelling protein coats) and another in radioactive phosphorus (32P, labelling DNA). When these phages infected E. coli, they shook the mixture in a blender to remove phage coats and centrifuged it. Radioactive 32P (DNA) was found inside the bacterial cells, while 35S (protein) stayed outside in the coats. Since only DNA entered the bacteria and directed the production of new phages, they proved that DNA — not protein — is the genetic material.

Q4. Describe elongation phase of DNA replication.

Elongation phase of DNA replication: after unwinding by helicase and stabilization by single-strand binding proteins, primase synthesizes a short RNA primer on each template strand. DNA polymerase III then adds nucleotides in the 5' to 3' direction: the leading strand is synthesized continuously toward the replication fork, while the lagging strand is synthesized discontinuously away from the fork as Okazaki fragments, each started by its own RNA primer. Later, DNA polymerase I removes the RNA primers and fills the gaps with DNA, and DNA ligase seals the nicks, producing two complete daughter strands. Synthesis of a new strand always begins with an RNA primer.

Q5. Describe molecular concept of a gene.

Molecular concept of a gene: a gene is a specific sequence of

nucleotides on DNA that codes for a functional product — a polypeptide or an RNA molecule (mRNA, tRNA or rRNA). The term gene was coined by Johannsen. A gene consists of a coding region (exons) and, in eukaryotes, non-coding regions (introns) removed during processing; it has a promoter region where RNA polymerase binds and a terminator region. One gene generally codes for one polypeptide (one gene-one polypeptide hypothesis, an extension of Beadle and Tatum's one gene-one enzyme hypothesis). Genes are located at specific loci on chromosomes; alternate forms of a gene are called alleles.

Q6. Describe the experiment of Fredric Griffith.

Griffith's experiment (1928): Frederick Griffith worked with two strains of Streptococcus pneumoniae — the virulent smooth (S) strain with a capsule, which killed mice, and the non-virulent rough (R) strain without a capsule, which did not. Heat-killed S bacteria alone did not kill mice, but a mixture of heat-killed S and live R bacteria killed the mice, and live S bacteria were recovered from them. Griffith concluded that a 'transforming principle' from the dead S bacteria had transformed live R bacteria into virulent S bacteria. Later Avery, MacLeod and McCarty (1944) identified the transforming principle as DNA.

Q7. Explain gene and genetic code.

A gene is a sequence of DNA nucleotides coding for a functional RNA or polypeptide. The genetic code is the set of rules by which the base sequence of mRNA is translated into amino acids: it is a triplet code, each codon (three nucleotides) coding one amino acid.

Its characteristics: it is universal (nearly the same in all organisms), specific (a codon codes a particular amino acid), degenerate (several codons can code the same amino acid, e.g. UUU and UUC for phenylalanine), and non-overlapping and commaless (codons are read in order without gaps). Of 64 codons, 61 are sense codons and 3 are stop codons (UAA, UAG, UGA); AUG is the initiation codon (methionine).

Q8. Describe post transcriptional modification of mRNA.

Post-transcriptional modification of mRNA (in eukaryotes): the primary transcript (pre-mRNA) is modified in the nucleus before leaving for the cytoplasm. (1) 5' capping — a modified guanine cap is added at the 5' end, protecting the mRNA and helping ribosome binding. (2) 3' polyadenylation — a tail of adenine nucleotides (poly-A tail) is added at the 3' end, increasing stability. (3) Splicing — introns (non-coding regions) are removed and exons (coding regions) are joined by the spliceosome. The mature mRNA then moves to the cytoplasm for translation.

Q9. Explain how prokaryotic and eukaryotic protein synthesis differ from each other.

Differences between prokaryotic and eukaryotic protein synthesis: (1) In prokaryotes, transcription and translation occur together in the cytoplasm (no nucleus), and mRNA needs no processing; in eukaryotes, transcription occurs in the nucleus, the mRNA is processed (capping, polyadenylation, splicing) and then translated in the cytoplasm. (2) Prokaryotic mRNA is often polycistronic (several genes on one mRNA), while eukaryotic mRNA is

monocistronic (one gene). (3) Prokaryotic translation starts with formyl-methionine and uses 70S ribosomes; eukaryotic translation starts with methionine and uses 80S ribosomes. (4) Prokaryotic genes lack introns; eukaryotic genes contain introns removed by splicing.

Q10. Write an essay on Down's syndrome.

Down's syndrome (Mongolism): an essay. Down's syndrome is a chromosomal disorder caused by trisomy of chromosome 21 — the person has 47 chromosomes (three copies of chromosome 21) instead of 46, due to non-disjunction of chromosome 21 during meiosis (failure of homologous chromosomes to separate), more common with increasing maternal age. Symptoms: flat (broad) face with slanting eyes, small ears, short stature, protruding tongue, single palmar crease, poor muscle tone, mental retardation, and congenital heart defects. There is no cure; affected children need medical care, special education and family support. It is an autosomal non-disjunction disorder, unlike sex-chromosome disorders such as Turner's (XO) and Klinefelter's (XXY) syndromes.

Unit 24: Evolution

Exercise - Section I: Objective Questions (MCQs)

Q1. Endosymbiont hypothesis explains origin of: (Mitochondria and chloroplasts originated as engulfed prokaryotes, giving rise to eukaryotic cells.)

- ✓ Eukaryotes
- Prokaryotes
- Armadillo
- Fish

Q2. The armoured mammal that live only in America is:

- Echidna
- Pangolin
- Porcupine
- ✓ Armadillo

Q3. Hardy-Weinberg theorem describes the frequencies of genotype of non-evolving:

- Family
- ✓ Population
- Species
- Community

Q4. Emigration and immigration of members of a population, cause disturbance in the: (Migration changes the gene pool by gene flow;

genetic drift is random change, not migration.)

Genotype

Phenotype

✓ Gene pool

Genetic drift

Q5. Analogous structure are: (Book Summary, p314: 'Analogous structures are similar in unrelated organisms' — similar in function, different in anatomy.)

Anatomically similar and functioning similarly

Anatomically similar but functioning differently

✓ Anatomically different but functioning similarly

Anatomically different and functionally differently

Q6. The theory of use and disuse was given by:

Darwin

✓ Lamarck

Aristotle

Mendel

Q7. Book 'The origin of species' was written by:

Linnaeus

Cuvier

Lyell

✓ Darwin

Q8. Lamarck theory of natural selection is usually known as:

Natural selection

✓ Inheritance of acquired characters

Descent with change

Mutation

Q9. The first eukaryotic organism appeared on earth about: (Book Summary, p314: 'Fossils record indicates that eukaryotes evolved from prokaryotes about 2 billion years ago'.)

1 billion years ago

One and half billion years ago

Three million years ago

✓ 2 billion years ago

Q10. Who figured out the mechanics of evolution known as natural selection? (Wallace independently conceived the same idea at about the same time, but the theory is attributed to Darwin.)

✓ Charles Darwin

Alfred Russell

Lamarck

Buffen

Q11. Genetic drift is change of gene frequency in: (NOTE: drift is also measured from one generation to the next, but its defining feature is random change by chance in small gene pools.)

Same generation

One generation to next

✓ By chance

Appearance of recessive allele

Q12. Species occurring in different geographical area are called as:

- Sympatric
- ✓ Allopatric
- Peripatric
- Parapatric

Q13. Which of the following is most important for speciation?

- Seasonal isolation
- ✓ Reproductive isolation
- Behavior isolation
- Tropical isolation

Q14. In the Hardy Weinberg equation, the frequency of dominant allele is represented by: (p + q = 1; p^2 + 2pq + q^2 = 1.)

- q^2
- q
- p^2
- ✓ p

Exercise - Section I: Fill in the blanks

The Vent hypothesis suggests that life may have begun at submarine ____ Vent.

hydrothermal

The study of the geographical distribution of fossil and living organisms is called ____.

biogeography

The fossil record of horse indicates that earliest horses had ____ toes.

four

There are two hypotheses about the evolution of eukaryotes from prokaryotes (i) membrane invagination theory and (ii) ____.

endosymbiont theory

Darwin presented his theory of natural selection in ____ in his famous book 'On the Origin of Species'.

1859

Darwin presented his theory of natural selection in 1859 in his famous book ____.

On the Origin of Species

Thomas Malthus was an English ____.

economist

Neo-Darwinism also called modern ____.

synthesis

Hardy Weinberg principle describes how a population can remain at genetic ____.

equilibrium

Genetic drift is a change in allele ____.

frequency

In parapatric speciation, the populations are not ____ by a physical barrier.

separated

Exercise - Section II: Short Questions

Q1. Define concept of special creation.

Special creation is the concept that all species were created by a supernatural being in their present form, separately and perfectly adapted, and have not changed since creation. It holds that each kind of organism appeared independently, not by descent from common ancestors, and denies gradual change over time. It is a religious/philosophical idea, not a scientific theory, since it cannot be tested by observation or experiment.

Q2. Briefly describe evidence of evolution from anatomy.

Evidence of evolution from anatomy: (1) Homologous structures — organs with the same basic anatomy but different functions (e.g. forelimbs of humans, bats, whales and birds) indicate descent from a common ancestor. (2) Analogous structures — organs with different anatomy but similar functions (e.g. wings of birds and insects) show adaptation to similar environments, not common ancestry. (3) Vestigial organs — reduced, functionless structures (e.g. appendix, tail bone in humans) are remnants of organs functional in ancestors.

Q3. Illustrate membrane invagination theory.

Membrane invagination theory: one hypothesis for the origin of

eukaryotes from prokaryotes. It proposes that the plasma membrane of an early prokaryotic cell folded inward (invaginated) to form the nuclear envelope, endoplasmic reticulum and other internal membranes, enclosing the DNA in a nucleus. The other hypothesis is the endosymbiont theory, in which mitochondria and chloroplasts originated from engulfed bacteria. Invagination explains the origin of the nucleus and endomembrane system, but not of mitochondria and chloroplasts.

Q4. Describe the drawback of Lamarckism.

Drawbacks of Lamarckism: (1) Acquired characters are not inherited — experiments (e.g. Weismann cutting the tails of mice for many generations) showed offspring still born with tails, since changes in body cells do not alter germ cells. (2) Use and disuse cannot explain all adaptations, e.g. the long neck of the giraffe or the vestigial organs, which do not result from use. (3) It has no mechanism for the origin of new variations, which modern genetics attributes to mutation and recombination.

Q5. Write the influence of Finches to Darwin.

Influence of finches to Darwin: during his voyage on HMS Beagle (1831, when Darwin was just 22 years old), Darwin observed finches on the Galapagos Islands. The finches on different islands had different beak shapes suited to different foods (seeds, insects, cactus), although they clearly descended from a common mainland ancestor. This suggested that species change over time in response to their environment — a key observation leading to natural selection.

Q6. Write the reasons for attributing theory of natural selection to Darwin.

Reasons for attributing the theory of natural selection to Darwin: (1) Darwin collected vast evidence during the Beagle voyage (1831) and over twenty years of study, including finches, fossils, armadillos and the geographical distribution of organisms. (2) He published the theory in full in 'On the Origin of Species' (1859), with evidence and mechanism, while Wallace's paper was shorter. (3) He explained the mechanism — variation, overproduction, struggle for existence and survival of the fittest — and was influenced by Malthus (population grows faster than resources), Lyell (uniformitarianism) and Hutton. Hence the theory is known as Darwinism, although Wallace reached the same idea independently.

Q7. Write factors that change allele frequency, briefly explain any one.

Factors that change allele frequency: mutation, natural selection, genetic drift, migration (gene flow) and non-random mating. Example — genetic drift: random changes in allele frequency in small gene pools due to sampling errors in the propagation of alleles; by chance, some alleles may be lost and others fixed, especially after a bottleneck (bottleneck effect) or when a few individuals found a new population (founder effect), even without any selective advantage.

Q8. Illustrate about bottleneck effect.

Bottleneck effect: a form of genetic drift in which a population is

drastically reduced in size by a disaster (e.g. disease, flood, hunting), so that only a few individuals survive. The survivors carry only a sample of the original gene pool, and some alleles are lost by chance; the rebuilt population has much less genetic variation, even if it grows large again. Example: cheetahs and northern elephant seals show very low genetic variation due to past bottlenecks.

Q9. Differentiate between allopatric and parapatric speciation.

Allopatric speciation: speciation in which populations are separated by a geographical (physical) barrier (e.g. a river, mountain or sea), so gene flow stops; the separated populations evolve independently under different conditions and eventually become reproductively isolated. Parapatric speciation: speciation in which populations are not separated by a physical barrier but occupy adjacent areas with different environments; gene flow is reduced (not fully stopped) by selection for different adaptations, and reproductive isolation gradually develops. The key difference is the presence of a physical barrier in allopatric speciation and its absence in parapatric speciation.

Q10. Describe the survival of the fittest.

Survival of the fittest: the principle that individuals best adapted to their environment are more likely to survive and reproduce, passing their favourable characters to the next generation, while less fit individuals die or leave fewer offspring. 'Fittest' means best fitted to the environment in terms of reproductive success, not merely physical strength. Over generations, this differential survival

changes allele frequencies and leads to adaptation and, eventually, new species.

Exercise - Section III: Extensive Questions

Q1. Describe Lamarck theory of evolution.

Lamarck's theory of evolution (Lamarckism) has two ideas: use and disuse of organs, and inheritance of acquired characters. Lamarck proposed that organs used more develop and become stronger, while organs not used weaken and disappear; and that characters acquired by an organism during its lifetime are passed to its offspring. Example: giraffes stretching their necks to reach high leaves developed long necks, which their offspring inherited; the ancestors of snakes lost their legs by disuse. Drawbacks: acquired characters are not inherited (Weismann's mice experiment), and the theory cannot explain vestigial organs or the origin of variations, which modern genetics attributes to mutation.

Q2. Explain the observations of Darwin during voyage, summarize them.

Darwin's observations during his voyage on HMS Beagle (1831, aged 22): (1) Fossils of extinct animals (e.g. giant ground sloths) resembled living animals in the same region, suggesting descent with change. (2) The armadillo, an armoured mammal living only in America, and related species differed between regions. (3) On the Galapagos Islands, finches and tortoises differed from island to island in beak shape and shell shape, although descended from mainland ancestors, showing adaptation to local conditions. (4) The geographical distribution of organisms (biogeography) showed

related species in nearby areas. Summary: species are not fixed; they change over time, and related species share common ancestors.

Q3. Explain Darwin's theory of natural selection.

Darwin's theory of natural selection: (1) Overproduction — organisms produce more offspring than can survive. (2) Variation — offspring vary, and some variations are heritable. (3) Struggle for existence — since resources are limited (Malthus: human population grows faster than the resources it depends on), organisms compete for food, space and mates. (4) Survival of the fittest — individuals with favourable variations survive and reproduce more, passing their characters on. (5) Descent with modification — over many generations, favourable characters accumulate and populations change, eventually forming new species. Natural selection is therefore differential survival and reproduction of the best-adapted individuals.

Q4. Describe Hardy Weinberg theorem and its conditions for stability?

The Hardy-Weinberg theorem describes how a population can remain at genetic equilibrium: allele and genotype frequencies stay constant from generation to generation if no evolution occurs. For two alleles with frequencies p (dominant) and q (recessive), p + q = 1 and the genotype frequencies are p^2 + 2pq + q^2 = 1 (p^2 homozygous dominant, 2pq heterozygous, q^2 homozygous recessive). Conditions for stability: (1) large population (no genetic drift), (2) no mutation, (3) no migration (no gene flow), (4) random

mating, and (5) no natural selection. If any condition is violated, allele frequencies change and evolution occurs.

Q5. Define speciation and explain its types.

Speciation is an evolutionary process by which a population undergoes change to such a degree that it becomes a new species — a group that can no longer interbreed with the original population (reproductive isolation). Its types: (1) Allopatric speciation — populations separated by a geographical barrier evolve independently into new species. (2) Sympatric speciation — new species arise within the same geographical area, e.g. by polyploidy in plants or by different food habits. (3) Parapatric speciation — populations in adjacent areas with different environments diverge without a physical barrier. (4) Peripatric speciation — a small peripheral population becomes isolated and evolves into a new species. Reproductive isolation (seasonal, behavioural, mechanical, gametic or hybrid sterility) is the most important factor in speciation.

Unit 25: Man and His Environment

Exercise - Section I: Objective Questions (MCQs)

Q1. Change in community structure of an ecosystem over a period of time: (Book Summary, p339: 'Ecological Succession is the gradual process by which ecosystem changes and develops over time'.)

- Niche
- Unstable ecosystem
- ✓ Succession
- Pioneer

Q2. Herbaceous stage in xerosere is the: (Book text, p328: xerosere stages are i) Crustose Lichen, ii) Foliose Lichen, iii) Moss, iv) Herbaceous, v) Shrub, vi) Tree and Climax.)

- First stage
- Third stage
- ✓ Fourth stage
- Last stage

Q3. The ozone layer has developed a hole over the:

- Arctic
- Equator
- ✓ Antarctica
- Tropics

Q4. As CFCs rise to the atmosphere, the ultraviolet rays release: (UV radiation breaks CFCs, releasing chlorine atoms that destroy ozone.)

- Fluorine
- ✓ Chlorine
- Carbon
- Nitrogen

Q5. The group of organisms that fix atmospheric nitrogen are:

- Plants
- ✓ Bacteria
- Fungi
- Insects

Q6. The producers of ecosystem are:

- Decomposers
- Absorptive heterotrophs
- Ingestive heterotrophs
- ✓ Autotrophs

Q7. Ozone layer is found in: (Book text, p333: 'The ozone layer is concentrated in 15-30 km, above the earth's surface in stratosphere'.)

- Troposphere
- ✓ Stratosphere
- Hydrosphere
- Mesosphere

Q8. The cause of greenhouse effect is:

- ✓ CO_2
- Hydrogen
- Nitrogen
- Oxygen

Q9. The graphical representation of ecological data of an ecosystem is called: (Book Summary, p339: 'The diagram that shows the relative amount of energy or matter or number of organisms within each trophic level in a food chain or food web is called pyramid'.)

- Succession
- Niche
- Habitat
- ✓ Pyramid

Q10. Which of these levels of ecological study involves both abiotic and biotic components:

- Organisms
- Population
- ✓ Ecosystem
- Community

Q11. Which of the following is renewable resource?

- Oil and natural gas
- Water and oil
- ✓ Air and water
- Oil and coal

Q12. About 95% of our daily energy requirements are fulfilled by: (Book value; fossil fuels dominate energy use.)

- Atomic energy
- Hydroelectric power
- ✓ Fossil fuel
- Wind energy

Q13. The ozone molecule is made up of by building of three atoms of:

- Nitrogen
- Hydrogen
- ✓ Oxygen
- Carbon

Exercise - Section I: Fill in the blanks

Every organism must get nutrients for its ____.

survival

The water cycle is driven by ____ energy.

solar

The total amount of solar energy which is fixed by the producers during photosynthesis is called ____.

gross primary productivity

Ecological succession which begins in water is called ____.

hydrosere

Increase in the number of individuals of a population is called ____ growth. (Book text, p329: 'Increase in the number of individuals of a population is called population growth'.)
population

The maximum population size that a particular environment can store is called ____ capacity. (Book Summary, p339: 'The carrying capacity is defined as the maximum population size that a particular environment can support'.)
carrying

The energy resources that have been used from long time is called ____ source of energy.
conventional

Acid rain may fall hundreds of miles away from the source of ____. (Book text, p333: 'The acid rain may fall hundreds of miles away from the source of pollution'.)
pollution

The Chernobyl disaster incidence took place on 26, April ____.
1986

Exercise - Section II: Short Questions

Q1. Describe trophic level.

A trophic level is a feeding level in a food chain or food web, showing the position of organisms in the flow of energy: the first

trophic level is producers (autotrophs, e.g. green plants), the second is primary consumers (herbivores), the third is secondary consumers (carnivores eating herbivores), and higher levels are tertiary consumers. A lot of energy is lost to the environment as food is transferred from one trophic level to the next, which is why food chains are short.

Q2. Define nitrification and role of bacteria in this process.
Nitrification is the conversion of ammonia/ammonium into nitrites and then nitrates by bacteria. The role of bacteria: Nitrosomonas bacteria oxidize ammonia (NH3) to nitrites (NO2-), and Nitrobacter bacteria oxidize nitrites to nitrates (NO3-), which plants can absorb and use to make proteins and nucleic acids. Nitrifying bacteria are chemosynthetic and obtain energy from these oxidations.

Q3. Differentiate between denitrification and nitrification.
Nitrification is the conversion of ammonia into nitrites and then nitrates by nitrifying bacteria (Nitrosomonas, Nitrobacter), making nitrogen available to plants. Denitrification is the opposite process: denitrifying bacteria (e.g. Pseudomonas) convert nitrates back into free nitrogen gas (N2), which returns to the atmosphere, removing nitrogen from the soil. Thus nitrification adds usable nitrogen to the ecosystem, while denitrification removes it.

Q4. Illustrate ecological pyramids.
Ecological pyramids are graphical representations of ecological data of an ecosystem, showing the relative amount of energy, biomass or number of organisms at each trophic level in a food

chain or food web. The three types are: pyramid of number (number of organisms at each level, usually upright but inverted for a tree ecosystem), pyramid of biomass (total dry mass at each level), and pyramid of energy (energy at each level, always upright because energy is lost at each transfer).

Q5. Write the reasons why food chains are shorter?

Food chains are shorter because: (1) A lot of energy is lost to the environment (as heat, in respiration, movement and waste) as food is transferred from one trophic level to the next — only about 10% passes on. (2) After 4-5 levels, so little energy remains that it cannot support another consumer level. (3) The number of organisms and biomass decrease at higher levels, so top consumers are few and cannot support further levels.

Q6. Write some causes of acid rain.

Causes of acid rain: (1) Burning of fossil fuels (coal, oil, natural gas) in power plants, industrial boilers and smelters releases sulphur dioxide (SO2). (2) Automobiles release nitrogen oxides (NOx). (3) Natural sources: volcanic eruptions and forest fires emit SO2 and NOx. These gases react with water vapour in the atmosphere to form sulphuric acid and nitric acid, which fall as wet deposition (rain, snow, fog, hail) or dry deposition (dust, gases). Human activities (anthropogenic sources) cause the majority of acid rain.

Q7. Describe briefly greenhouse effect.

Greenhouse effect: the process by which greenhouse gases (CO2, methane, water vapour) in the atmosphere trap heat radiated from

the Earth. The Sun's radiation reaches the Earth; some energy is reflected and the rest absorbed by the Earth's surface; the Earth emits heat back toward space, but greenhouse gases trap most of it, keeping the planet warm. The amount of CO2 in the air is increasing day by day (about 0.04% now, compared with 0.03% in the mid-twentieth century), causing global warming — rising average temperatures, melting polar ice, sea-level rise and changing rainfall patterns.

Q8. Differentiate between industrial and biological fixation of nitrogen.

Industrial fixation of nitrogen is the artificial conversion of atmospheric nitrogen into ammonia by the Haber process, using high temperature and pressure with a catalyst, for fertilizers. Biological fixation of nitrogen is the natural conversion of atmospheric nitrogen into ammonia by nitrogen-fixing bacteria (e.g. Rhizobium in legume root nodules, free-living Azotobacter, cyanobacteria), using the enzyme nitrogenase. Industrial fixation is man-made, energy-intensive and large-scale; biological fixation is natural, enzyme-driven and occurs in soil and root nodules.

Q9. Differentiate between hydrosere and xerosere.

Hydrosere is ecological succession beginning in water (ponds, lakes, marshes or elsewhere in open water), where plants can grow — stages pass from submerged and floating plants to reed swamp, marsh and finally woodland. Xerosere is ecological succession beginning in dry areas (bare rocks, sand dunes, rocky slopes) — stages pass from crustose lichen, foliose lichen, moss, herbaceous,

shrub to tree and climax forest. Hydrosere starts wet and becomes drier; xerosere starts extremely xeric and becomes moister as soil and humus accumulate.

Q10. Define the following terms. i) Aquifers ii) Net primary production iii) Climax community.

Aquifers are underground layers of water-bearing permeable rock (e.g. sandstone) from which groundwater can be extracted through wells; they store water that percolates down from the surface. Net primary production is the amount of energy (or biomass) left after producers use part of the gross primary productivity for their own respiration — i.e. gross primary productivity minus respiration — available to consumers. Climax community is the final, stable community of ecological succession (e.g. climax forest with thick growth of mesophytic trees with characteristic under-shrubs and herbs), which remains stable as long as the environment is unchanged.

Q11. Briefly describe role of NGOs to tackle environmental problems.

Roles of NGOs in tackling environmental problems: NGOs (e.g. IUCN — International Union for Conservation of Nature, and WWF — World Wide Fund for Nature) create awareness about environmental problems (deforestation, air pollution, climate change, pesticides misuse, desertification), run conservation projects (protection of forests and wildlife), pressurize governments to implement environmental acts, help in reforestation and clean-up campaigns, and provide expertise and funds. In Pakistan, NGOs work with the

Forest Department and the Antipollution Department to protect the environment.

Q12. Differentiate between renewable and non-renewable energy resources.

Renewable energy resources are those that are replenished naturally and will not run out, e.g. air (wind), water (hydel power), solar energy and biomass; they cause little pollution.

Non-renewable energy resources are those that form at a much slower rate than their consumption and will eventually be exhausted, e.g. fossil fuels (coal, oil, natural gas) and nuclear fuels; about 95% of our daily energy requirements are fulfilled by fossil fuels. Non-renewable resources also cause pollution (CO_2, acid rain), while renewable resources are cleaner.

Q13. Describe effects of ozone layer depletion.

Effects of ozone layer depletion: (1) More ultraviolet radiation reaches the Earth, increasing skin cancer and eye cataracts in humans. (2) It damages crops and reduces agricultural yields. (3) It harms aquatic life, especially plankton, disturbing food chains. (4) It weakens the immune system of humans and animals. (5) It damages materials such as plastics and paints. The ozone hole over Antarctica developed because chlorofluorocarbons (CFCs), commercially important compounds in the stratosphere, release chlorine atoms under UV rays, which destroy ozone molecules.

Exercise - Section III: Extensive Questions

Q1. Describe water cycle.

The water cycle is driven by solar energy. Water evaporates from oceans, lakes and land and transpires from plants, rising as water vapour into the atmosphere. The vapour cools and condenses into clouds, and falls back as precipitation (rain, snow, hail). Part of the water runs off into rivers and seas, part infiltrates into the soil and collects in aquifers as groundwater, and part is taken up by plants and returned by transpiration. Thus water circulates continuously between the Earth's surface and the atmosphere, powered by the Sun's energy.

Q2. Describe different components of ecosystem.

Components of an ecosystem: (1) Abiotic components — non-living factors: light, temperature, water, air, soil and minerals. (2) Biotic components — living factors: producers (autotrophs, green plants and algae) that fix solar energy; consumers (heterotrophs) — primary consumers (herbivores), secondary consumers (carnivores) and tertiary consumers; and decomposers (bacteria, fungi) that break down dead matter and return nutrients to the soil. Energy flows from producers to consumers and decomposers, while nutrients cycle through biogeochemical cycles (water, carbon, nitrogen).

Q3. Define ecological pyramid and explain pyramid of energy biomass and number.

An ecological pyramid is the graphical representation of ecological data of an ecosystem, showing the relative amount of energy, biomass or number of organisms at each trophic level in a food chain or food web. Pyramid of energy: always upright, because

energy decreases at each level as much is lost as heat. Pyramid of biomass: usually upright (total dry mass decreases upward), but inverted in some aquatic ecosystems where producers (phytoplankton) have less biomass than consumers at a given moment. Pyramid of number: usually upright (many producers, fewer consumers), but inverted when one large tree supports many herbivores and parasites.

Q4. Define succession. Explain two types of ecological succession.
Succession is the gradual process by which an ecosystem changes and develops over time, with one community replacing another. Its two types: (1) Primary succession — succession on a bare area where no soil exists (e.g. bare rock, new volcanic island), beginning with pioneer species (lichens). (2) Secondary succession — succession on an area where a community previously existed but was destroyed (e.g. after fire or flood), where soil already exists, so it is faster. By habitat, succession is hydrosere (begins in water), mesosere (sufficient moisture) or xerosere (dry areas). Succession ends in a stable climax community.

Q5. Explain the characteristics of population dynamics?
Population dynamics is the branch of life science concerned with the long-term and short-term changes in size, age and composition of a population. Its characteristics: (1) Growth — increase in the number of individuals (lag phase, then exponential/geometric growth as the population doubles at regular intervals). (2) Density — number of individuals per unit area. (3) Distribution — how individuals are spread in an area. (4) Carrying capacity — the

maximum population size that a particular environment can support. (5) Viable size — the minimum population needed for survival. Populations are affected by birth and death rates and by immigration and emigration.

Q6. Define nuclear energy and explain its advantages and disadvantages.

Nuclear energy is the energy released from the nucleus of an atom, by fission (splitting heavy nuclei, e.g. uranium) or fusion, used to generate electricity and heat. Advantages: it produces a large amount of energy from a small amount of fuel, does not release CO2 or smoke (so it does not cause air pollution or global warming), and uranium reserves last long. Disadvantages: it produces dangerous radioactive waste that remains hazardous for thousands of years; accidents (e.g. Chernobyl, 26 April 1986) release deadly radiation; nuclear plants are expensive to build and decommission; and nuclear material can be misused for weapons.

Q7. Define ozone layer and describe its composition, formation, role and causes of its depletion.

The ozone layer is the region of the stratosphere (concentrated 15-30 km above the Earth's surface) rich in ozone (O3, a molecule of three oxygen atoms), which protects all life from the Sun's harmful ultraviolet radiations. Formation: UV radiation splits oxygen molecules (O2) into atoms, which combine with O2 to form O3. Role: it absorbs most UV-B and UV-C radiation, preventing skin cancer, cataracts and damage to crops and plankton. Causes of depletion: chlorofluorocarbons (CFCs), commercially important

compounds in the stratosphere, release chlorine atoms under UV rays, which break down ozone; the ozone hole has developed over Antarctica. Effects include more skin cancer, eye damage, reduced crop yields and harm to aquatic life.

Q8. Differentiate between conventional and non-conventional energy resources, Discuss them in detail.

Conventional energy resources are those used for a long time: fossil fuels (coal, oil, natural gas) and wood; they are non-renewable, form much slower than their consumption, and cause pollution (CO_2, acid rain, global warming). About 95% of daily energy requirements are fulfilled by fossil fuels. Non-conventional energy resources are newer, renewable alternatives: solar energy, wind energy, hydel (water) power, geothermal energy, tidal energy and nuclear energy. They are cleaner and inexhaustible (except nuclear fuel), but need higher initial investment and technology. Pakistan faces environmental problems (deforestation, air pollution, climate change, desertification) partly from fossil-fuel use, so non-conventional sources are increasingly important.

Q9. Explain different stages of xerosere succession.

Stages of xerosere succession (xerarch succession on bare rock): i) Crustose lichen stage — pioneer lichens (e.g. Rinodina, Lecidea) flourish during short moist periods and withstand long drought, accumulating organic matter and dust. ii) Foliose lichen stage — crustose lichens are replaced by foliose lichens (e.g. Parmelia), which absorb more water and build more organic matter, providing a base for mosses. iii) Moss stage — xerophytic mosses (e.g.

Polytrichum, Tortula) occupy the soil, absorb more water through rhizoids and form humus supporting herbs. iv) Herbaceous stage — annual herbs invade, then biennial and perennial herbs, enriching the soil and increasing its moisture. v) Shrub stage — shrubs invade and overshadow herbs, and the soil becomes richer with greater water-retaining capacity. vi) Tree and climax stage — xerophytic trees invade, then mesophytic trees replace them, forming a climax forest with characteristic under-shrubs and herbs.

Unit 26: Biotechnology

Exercise - Section I: Objective Questions (MCQs)

Q1. The enzyme which joints two pieces of DNA is:

- DNA polymerase III
- Restriction endonuclease
- ✓ DNA ligase
- DNA polymerase I

Q2. Antibody made by soybeans can be used as treatment for: (FLAGGED: the soybean-produced plantibody targets herpes simplex virus type 2, which causes genital herpes; 'Herpes Simplex' names the same virus.)

- AIDS
- Herpes Simplex
- ✓ Genital Herpes
- Hepatitis

Q3. Antithrombin III is biotechnical product produced in: (Antithrombin III (hAT) is the classic transpharmer product of transgenic goats, secreted in milk; the book's text, p371, names the protein but not the animal, so this follows the standard product-animal pairing.)

- Sheep
- ✓ Goat
- Mice

Cow

Q4. Cell suspension cultures of Digitalis lanata produce.

Anti-toxin

✓ Digitoxin

Polludrin

Quinine

Q5. EcoRI is commonly used as: (EcoRI is a restriction endonuclease from E. coli.)

Bacteriophage

Gene

✓ Restriction enzymes

DNAase

Q6. Recombinant DNA is introduced into the host cell by means of:

Phage

✓ Vector

Bacterium

Fungus

Q7. The PCR was developed by K. Mullis in:

1970

✓ 1983

1975

1978

Q8. Taq polymerase is an enzyme present in: (Taq polymerase

comes from Thermus aquaticus, a bacterium living in hot springs, and survives the high temperatures of PCR.)

✓ Bacteria

Protozoans

Algae

Helminths

Q9. A complete set of genes of an individual is called as:

Gene pool

✓ Genome

Gene library

Recombinant gene

Q10. Those organisms which have had a foreign gene into them are called as:

✓ Transgenic

Transmuted

Hermaphrodites

Polygenesis

Q11. The use of transgenic animal to produce pharmaceutical is termed as:

✓ Gene pharming

Antibiotic

Gene therapy

Antiviral

Q12. Transgenic soybeans are made to resist against:

(Herbicide-resistant soybeans, e.g. Roundup Ready, are the classic example.)

✓ Herbicides

Fungicides

Insecticides

Pesticide

Q13. Which of the following is the genetic marker that is used in DNA finger printing.

Primer

Probe

✓ RFLP

Intron

Q14. RFLP is a (an): (FLAGGED: RFLP — restriction fragment length polymorphism — is a variation in restriction-fragment lengths, typically in non-coding DNA; intron is the closest option among those printed.)

✓ Intron

Exon

Anticodon

Codon

Exercise - Section I: Fill in the Blanks

Recombinant DNA technology is a series of procedures that is used to join ____ Segments.

DNA

First restriction enzyme was discovered in 1970, by ____.

Hamilton Smith

The collection of bacterial clone containing recombinant DNA is called ____ library.

genomic

Human genome comprises ____ billion nucleotides.

3

The initial plant part which is used to develop tissue culture is called ____.

explant

Interferons are used to treat ____ infections.

viral

A plant, containing foreign gene is called ____.

transgenic

The transgenic animals which produce drugs are also called ____ animals. (Book text, p371: 'such transgenic animals are called transpharmer animals'.)

transpharmer

DNA molecules with complementary sticky ends associated by ____ bond.

hydrogen

PCR is the abbreviation of ____.

Polymerase Chain Reaction

Exercise - Section II: Short Questions

Q1. Write the role of restriction endonucleases in gene cloning.

Restriction endonucleases (restriction enzymes) cut DNA at specific recognition sequences, producing sticky ends (or blunt ends). In gene cloning they are used to cut both the DNA containing the gene of interest and the vector (e.g. plasmid) with the same enzyme, so that the complementary sticky ends can join and DNA ligase seals them into recombinant DNA. EcoRI is a commonly used restriction enzyme. Without restriction enzymes, the gene of interest could not be isolated or inserted into a vector.

Q2. Write note on Taq polymerase.

Taq polymerase is a heat-stable DNA polymerase obtained from the bacterium Thermus aquaticus, which lives in hot springs. It is the key enzyme of PCR because it survives the repeated high-temperature denaturation steps (about 94-95 C) that would destroy ordinary DNA polymerases, and it synthesizes new DNA strands from primers during each cycle, allowing million-fold amplification of the target DNA.

Q3. Briefly describe monoclonal antibodies.

Monoclonal antibodies (mAB) are a group of identical antibodies made by identical immune cells, all clones of a unique parent cell. They are typically made by fusing myeloma cells (cancerous

B-lymphocytes) with spleen cells from a mouse immunized with the desired antigen — somatic cell hybridization. Because they bind specifically to one substance, they are used to detect or purify that substance, diagnose diseases (including herpes virus, polio, hepatitis), detect drugs and abnormal substances in blood, and separate a pure antigen for vaccine development.

Q4. Write short note on callus culture.

Callus culture is a plant tissue-culture technique in which an explant (initial plant part) is grown on a nutrient medium to form a callus — an undifferentiated mass of cells. The callus can be maintained by subculture and, by changing the hormones in the medium, induced to form shoots and roots (organogenesis) or somatic embryos, regenerating whole plantlets. Callus culture is used for micropropagation, production of secondary metabolites (e.g. digitoxin from Digitalis lanata cell suspensions) and creating somaclonal variation.

Q5. Define anther, explain briefly anther culture.

Anther is the pollen-producing part of the stamen of a flower, containing pollen grains (male gametophytes). Anther culture is a tissue-culture technique in which immature anthers (or isolated pollen) are cultured on a nutrient medium to produce haploid plants directly from pollen grains. Haploid plants can be chromosome-doubled to produce homozygous diploid plants in one generation, greatly speeding plant breeding compared with conventional inbreeding.

Q6. List some applications of animal cell culture.

Applications of animal cell culture: (1) Production of vaccines and viruses for study. (2) Production of pharmaceuticals and therapeutic proteins (e.g. interferons, hormones) by transpharmer animals and cultured cells. (3) Gene therapy research and testing of drugs and cosmetics without animal experiments. (4) Production of monoclonal antibodies by hybridoma cells. (5) Study of cell physiology, cancer and virus replication.

Q7. Write methods of gene transformation in plants.

Methods of gene transformation in plants: (1) Electroporation — protoplasts (plant cells with walls removed) are treated with an electric current in a liquid containing foreign DNA; the current makes tiny self-sealing holes in the plasma membrane through which the DNA enters. (2) Agrobacterium-mediated transformation — foreign DNA is inserted into the tumour-inducing (Ti) plasmid of Agrobacterium tumefaciens, the natural genetic engineer of plants; the bacterium transfers its T-DNA into the plant genome at wound sites. (3) Particle gun (biolistics) — DNA-coated microscopic metal particles (gold or tungsten) are bombarded into plant tissue (callus) by the particle gun developed by John C. Sanford and Theodore M. Klein (1987); somatic embryos develop into transgenic plants.

Q8. Define DNA/RNA probes.

DNA/RNA probes are fluorescent or radioactive labelled pieces of DNA or RNA, usually 100-1000 bases long, complementary to the sequence in the target DNA. When mixed with denatured target DNA, the probe hybridizes (binds) to its complementary sequence,

revealing its presence. Probes are widely used in diagnosis of microbial diseases (e.g. ready-made DNA probes for herpes virus, polio, hepatitis), detection of prenatal diseases (e.g. short tandem repeats, STR) and in DNA fingerprinting.

Q9. Illustrate bio-fertilizers.

Bio-fertilizers are living microorganisms (bacteria, cyanobacteria, fungi) used to enrich soil fertility instead of chemical fertilizers. Examples: Rhizobium bacteria in legume root nodules fix atmospheric nitrogen; Azotobacter and Azospirillum are free-living nitrogen fixers; mycorrhizal fungi help plants absorb phosphorus; cyanobacteria (e.g. Anabaena) fix nitrogen in rice fields. Bio-fertilizers are cheap, eco-friendly and do not pollute soil and water like chemical fertilizers.

Q10. Write brief note on nanotechnology.

Nanotechnology is the technology of manipulating matter at the nanometre scale (1-100 nm, i.e. at the level of atoms and molecules) to create new materials and devices with novel properties. In biology and medicine it is used for targeted drug delivery (nanoparticles carrying drugs directly to diseased cells), biosensors and biochips for rapid diagnosis, and improved imaging. Nanotechnology is an emerging field of biotechnology with applications in health, agriculture, industry and environment.

Q11. Explain Hazards and ethical implications of using biotechnology.

Hazards and ethical implications of using biotechnology: (1)

Biosafety — release of GMOs may disturb ecosystems (e.g. transgenic bacteria with antibiotic resistance or cancer genes, transgenic fish escaping to the wild). (2) Health risks — toxicity or allergy associated with microbial products and GM foods. (3) Biological warfare — genetic manipulation could be misused to produce biological weapons (most countries signed the Biological Weapons Convention of 1972 pledging never to do so). (4) Ethical, legal and social implications (ELSI) — concerns about 'playing God', patenting of life (intellectual property rights), and unequal access to benefits. (5) Contamination, infection or mutation of production strains must be prevented.

Q12. List some biotechnological products.

Biotechnological products include: insulin (humulin) from transgenic bacteria; human growth hormone; interferons for viral infections; vaccines (hepatitis B vaccine); monoclonal antibodies for diagnosis; Antithrombin III (hAT) and human lactoferrin (HLF) from transpharmer animals; herbicide-resistant and insect-resistant crops (e.g. Bt cotton, transgenic soybeans); bio-fertilizers; industrial enzymes; and DNA probes for disease diagnosis.

Q13. Write the process of the automated DNA sequencing.

Automated DNA sequencing (based on the Sanger-Coulson method): DNA fragments are produced by chain termination using fluorescently labelled dideoxynucleotides (each of the four ddNTPs carrying a different colour). The fragments are separated by size in a single capillary by gel electrophoresis, and a laser detector reads the colours in order as fragments pass, with a computer recording

the base sequence automatically. Automation allows whole genomes (e.g. the human genome, about 3 billion nucleotides) to be sequenced rapidly and accurately.

Q14. Write short note about biochips.

Biochips are miniaturized devices (microarrays) carrying thousands of tiny spots of DNA, proteins or other biological molecules on a solid surface (glass or silicon). A sample (e.g. patient DNA or RNA) is applied, and hybridization or binding at specific spots reveals which genes are present or active. Biochips are used for rapid diagnosis of diseases, detection of mutations, gene-expression studies and drug screening, and are an application of nanotechnology in biotechnology.

Exercise - Section III: Extensive Questions

Q1. Explain method for production of recombinant DNA.

Method for production of recombinant DNA: (1) Selection and isolation of the gene of interest — the desired DNA segment is obtained from donor DNA. (2) Cutting with restriction endonucleases — both the donor DNA and the vector (plasmid or phage) are cut with the same restriction enzyme, producing complementary sticky ends. (3) Integration of the DNA insert into the vector — the gene is inserted into the vector, and DNA ligase joins the pieces (DNA molecules with complementary sticky ends associate by hydrogen bonds, then ligase seals them). (4) Introduction into the host cell — the recombinant DNA is introduced into host bacteria by transformation, using a vector as carrier. (5) Selection of transformed cells — cells that took up the recombinant

DNA are identified (e.g. by antibiotic resistance or blue-white screening) and cultured; the collection of bacterial clones containing recombinant DNA is called a genomic library.

Q2. Describe uses of transgenic bacteria.

Uses of transgenic bacteria: (1) Production of human proteins — insulin (humulin), human growth hormone and interferons for treatment of diseases. (2) Production of vaccines and antigens for immunization. (3) Extraction of metals — bacteria are used to obtain copper, gold and uranium from low-grade ores, and genetically engineered organisms with improved bioleaching capabilities are being tested. (4) Degradation of pollutants and oil spills (bioremediation). (5) Production of industrial enzymes and chemicals. Ecological concerns: transgenic bacteria may create hazards if cancer-cell genes or antibiotic resistance spread, or if they disturb the natural environment when their numbers exceed limits.

Q3. Write note on components of PCR technique.

Components of PCR technique: (1) Template DNA — the DNA segment to be amplified. (2) Primers — short synthetic DNA pieces complementary to the ends of the target sequence, starting synthesis. (3) Taq polymerase — the heat-stable DNA polymerase from Thermus aquaticus that synthesizes new strands. (4) Deoxyribonucleotides (dNTPs) — the building blocks for new DNA. (5) Buffer with Mg2+ ions — maintaining pH and enzyme activity. The PCR cycle has three steps repeated about 30 times: denaturation (about 94 C, strands separate), annealing (primers

bind, about 55 C) and extension (Taq polymerase synthesizes, about 72 C), giving million-fold amplification. PCR is used for diagnosis (HBV, HCV, HIV detection), DNA fingerprinting, genome analysis and detection of microorganisms in food, water and environment.

Q4. Define genomic library and also explain it.

A genomic library is the collection of bacterial clones containing recombinant DNA that together represents the complete genome of an organism. It is made by cutting the organism's total DNA with restriction enzymes into fragments, inserting each fragment into vectors (plasmids or phages) with DNA ligase, and introducing the recombinant vectors into host bacteria; each bacterial clone carries one fragment. Screening the library with a DNA/RNA probe identifies the clone carrying the gene of interest. Genomic libraries are used to isolate genes, study genome organization and construct genome maps.

Q5. Write a detail note on gel electrophoresis.

Gel electrophoresis is a technique to separate DNA fragments by size. DNA samples are loaded into wells in an agarose gel and an electric current is applied: since DNA is negatively charged, fragments move toward the positive electrode, smaller fragments moving faster and farther than larger ones. After staining (e.g. with ethidium bromide) and viewing under UV light, bands of fragments appear. It is used in gene sequencing (Sanger method, where fragments differing by one base are separated), DNA fingerprinting (RFLP analysis), and checking the results of restriction digestion

and PCR.

Q6. Explain Sanger's methods of DNA sequencing techniques.

Sanger's method of DNA sequencing (Sanger-Coulson, chain-termination method): the DNA to be sequenced is used as a template for synthesis in four separate reactions, each containing normal dNTPs plus a small amount of one dideoxynucleotide (ddATP, ddTTP, ddGTP or ddCTP) that stops chain growth when incorporated. This produces sets of fragments ending at every occurrence of that base. The fragments are separated by size on a gel (or in a capillary in automated sequencing, with fluorescent labels), and reading the band order gives the base sequence. The method was used for the Human Genome Project.

Q7. Describe automated DNA sequencing technique.

Automated DNA sequencing technique: based on the Sanger-Coulson method, but with fluorescently labelled dideoxynucleotides (a different colour for each base) in a single reaction. The terminated fragments are separated by size in one capillary by gel electrophoresis, and as each fragment passes a laser detector, its colour is recorded by a computer, giving the sequence automatically and rapidly. Automation made large projects possible, including the Human Genome Project (James Watson as its first director), which sequenced the human genome of about 3 billion nucleotides and constructed genome maps and genetic markers.

Q8. Write note on Human genome project. What are its

applications?

The Human Genome Project (HGP) was the international project to determine the complete sequence of the human genome (about 3 billion nucleotides), identify all human genes, and construct genome maps. James Watson was its first director. Its goals: sequence the human genome, map genes and genetic markers, study variation, and develop tools for data analysis. Applications: diagnosis and treatment of genetic diseases (e.g. cystic fibrosis, sickle cell anaemia), gene therapy, understanding human evolution, forensic DNA fingerprinting, and personalized medicine. It also raised ethical, legal and social implications (ELSI) about privacy and use of genetic information.

Q9. Define cystic fibrosis, and also describe role of gene therapy in cystic fibrosis.

Cystic fibrosis (CF) is an inherited disease affecting the mucus and sweat glands: the affected person with severe symptoms has serious lung and digestive problems, while a mild form shows symptoms after adolescence. The cause is a defect in the Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) gene, which encodes a protein controlling the movement of salt and water in and out of cells; the defective gene makes the cells lining the lungs, oviduct, pancreas and other organs produce abnormally thick, sticky mucus, obstructing airways, ducts and glands. Role of gene therapy in CF: the CFTR genes were identified in 1989 (named CFTR); gene therapy creates a healthy version of the flawed CFTR gene and delivers it (e.g. by a harmless virus, AVV, dripped into the lungs through the nose) into lung cells, so the cells gain normal

genes and function normally. Approaches include inserting a normal gene to replace the non-functional gene, swapping the abnormal gene by homologous recombination, or altering gene regulation.

Q10. Explain the scope and importance of biotechnology in promoting human welfare.

Scope and importance of biotechnology in promoting human welfare: (1) Healthcare — production of insulin, growth hormone, interferons, vaccines (e.g. hepatitis B), monoclonal antibodies for diagnosis, and gene therapy for diseases like cystic fibrosis. (2) Agriculture — transgenic plants (first tobacco plants, 1986; herbicide-resistant soybeans, Bt crops) with higher yield and resistance. (3) Industry — enzymes, bio-fertilizers, bioleaching of metals (copper, gold, uranium). (4) Environment — bioremediation and conservation (IUCN, WWF). (5) Forensic science — DNA fingerprinting in crime and law. (6) Research — genome projects, genome maps, DNA sequencing. Biotechnology thus improves health, food production, industry and environment, while its hazards (biosafety, GMOs, biological warfare, intellectual property) must be managed.

Unit 27: Biology and Human Welfare

Exercise - Section I: Objective Questions (MCQs)

Q1. Td-vaccination is the best way to prevent a tetanus infection. (PRINT DEFECT — FLAGGED: as printed, this stem does not match the options. The statement itself is true (Td vaccination prevents tetanus), but none of the options answers it. The options belong to a 'study of ...' question, most plausibly 'the study of tissues is called', whose answer is Histology; that pairing is used here and marked uncertain.)

- Morphology
- ✓ Histology
- Anatomy
- Physiology

Q2. Casein is?

- ✓ Milk protein
- Starter culture
- Toxin
- Gyle

Q3. Integrated disease management aims to improve the quality of:

- Environment
- ✓ Life
- Food
- Animals

Q4. Tetanus is also called: (Tetanus, caused by Clostridium tetani, causes sustained muscle spasms, especially jaw stiffness — lockjaw.)

- Muscle cramp
- Muscle fatigue
- ✓ Lockjaw
- Paralysis

Q5. Animal husbandry is the branch of science deals with care of: (Book Summary, p395: 'Livestock are the domesticated animals reared commercially'.)

- ✓ Livestock
- Fruits
- Diseases
- Vaccination

Q6. The best way to prevent polio is:

- Antibiotics
- Cleanliness
- ✓ Vaccination
- Healthy food

Q7. Which of the following method is used for crop improvement:

- ✓ Selection
- Fertilization
- Vaccination
- Watering

Q8. The main component of sewage is:

- Oil
- Inorganic matter
- Minerals
- ✓ Organic matter

Q9. A back cross is made when a hybrid is crossed with its: (Book Summary, p395: 'Backcross is formed when a hybrid is crossed with one of his parents'.)

- Sibling
- Offspring
- ✓ Parents
- Other species

Exercise - Section I: Fill in the Blanks

Integrated disease management aims to improve and maintain the ____ of life.

quality

Vaccine is a biological preparation contain ____ in killed or attenuated form.

microorganisms

Prevention is better than ____.

cure

Hepatitis is an inflammatory condition of ____.

liver

About 30-40% of income in our country comes from ____.

agriculture

Hybridization does not change ____ contents of organism but it produces new combination of genes.

gene

Measles and mumps are caused by different viruses but share unique ____.

symptoms

A home garden creates a ____ environment in house.

healthy

Biofertilizers consist of living ____.

microorganisms

Microbes have tremendous potential to produce ____. (Book Summary, p395: 'Microbes have tremendous potential to produce energy'.)

energy

Exercise - Section II: Short Questions

Q1. Define integrated disease management.

Integrated disease management (IDM) is a strategy that combines all suitable methods — vaccination, hygiene, clean water, proper

nutrition, vector control, early diagnosis and treatment, and public awareness — to improve and maintain the quality of life by preventing and controlling diseases, rather than relying on a single method. It aims to reduce disease incidence at low cost with minimum side effects.

Q2. Write importance of vaccination against prevention of diseases.

Importance of vaccination against prevention of diseases: vaccination prepares the immune system by exposing it to a killed or attenuated microorganism (or its antigen), so that on real infection the body responds rapidly with antibodies and memory cells. It prevents diseases (e.g. polio, tetanus, measles), reduces epidemics and deaths, protects unvaccinated people through herd immunity, and is cheaper than treatment — prevention is better than cure. Td-vaccination is the best way to prevent tetanus infection, and vaccination is the best way to prevent polio.

Q3. Write two types of vaccines recommended for polio.

Two types of vaccines recommended for polio: (1) Oral polio vaccine (OPV, Sabin vaccine) — live attenuated poliovirus given by mouth, easy to administer and giving gut immunity. (2) Inactivated polio vaccine (IPV, Salk vaccine) — killed poliovirus given by injection, safe for immunocompromised persons. Both are used in polio eradication programmes.

Q4. Write importance of acclimatization for plant breeding.

Importance of acclimatization for plant breeding: acclimatization is the adaptation of a plant variety to a new environment over

generations. It is important in plant breeding because a variety bred elsewhere must adjust to local climate, soil, pests and diseases before it can yield well; acclimatized varieties show stable performance, and breeders select plants that have become adapted to local conditions for further improvement.

Q5. Define back cross, write its importance.

Backcross is a cross in which a hybrid (F1) is crossed with one of its parents (usually the recurrent, better parent). Its importance: it transfers a desired character (e.g. disease resistance) from a donor parent into an otherwise good variety, while recovering most of the recurrent parent's characters; repeated backcrossing produces a variety nearly identical to the recurrent parent plus the new trait. It is a standard method of crop improvement along with selection and hybridization.

Q6. Write importance of home gardening.

Importance of home gardening: a home garden creates a healthy environment in the house; it provides fresh vegetables, fruits and herbs for the family, improving nutrition at low cost; it gives physical exercise and recreation; it improves air quality and beauty of the surroundings; and it teaches children about plants and nature. Home gardening also reduces dependence on market produce and pesticides.

Q7. Write role of microbes for energy production.

Role of microbes for energy production: microbes have tremendous potential to produce energy. Anaerobic bacteria produce biogas

(mainly methane) by digesting sewage and animal dung in biogas plants; yeast produces ethanol by fermentation of sugars, used as biofuel; algae and bacteria can produce hydrogen and biodiesel; and microbes decompose organic waste in sewage treatment, releasing energy-rich gases. Microbial energy is renewable and reduces pollution.

Exercise - Section III: Extensive Questions

Q1. Define and explain animal husbandry and role of livestock in national economy.

Animal husbandry is the branch of science that deals with the care of livestock — domesticated animals reared commercially for sale on a farm or agricultural business. It includes proper feeding, watering and cleaning the living areas of animals, breeding for better traits, prevention and treatment of diseases (vaccination, hygiene), and management of milk, meat, wool and egg production. Role of livestock in the national economy: livestock contributes about 30-40% of income in Pakistan's agricultural sector (the book's figure for income from agriculture-related activity), provides milk, meat, eggs, wool, leather and draught power, gives employment to rural people, and earns foreign exchange through exports. Healthy livestock thus directly supports food security and the national economy.

Q2. Explain the latest techniques applied to enhance crop production.

Latest techniques applied to enhance crop production: (1) Selection — choosing the best plants for breeding (the basic method of crop

improvement). (2) Hybridization — crossing two different varieties to combine desired characters, producing hybrid vigour. (3) Backcrossing — crossing a hybrid with a parent to transfer a specific trait. (4) Genetic engineering and biotechnology — producing transgenic plants (e.g. herbicide-resistant soybeans, Bt crops) and using tissue culture for rapid micropropagation. (5) Bio-fertilizers (living microorganisms like Rhizobium) and integrated disease management to protect crops with less chemical use. (6) Acclimatization of varieties to local conditions.

Q3. Describe the role of microbes in human welfare.

Role of microbes in human welfare: (1) Food — microbes make yoghurt, cheese (casein is the milk protein), bread, pickles and alcoholic drinks; industries use microbes to make products valuable to man. (2) Medicine — antibiotics (e.g. penicillin), vaccines, vitamins and interferons are microbial products. (3) Agriculture — bio-fertilizers (Rhizobium, Azotobacter) fix nitrogen and enrich soil. (4) Industry — microbes produce enzymes, organic acids and alcohol. (5) Environment — microbes treat sewage and decompose waste, and produce energy (biogas, ethanol). (6) Research — microbes like E. coli are used in genetic engineering and biotechnology. Living organisms not visible to the naked eye are called microbes, and they are essential partners of human life.

Q4. Describe three steps of sewage treatment by microbes.

Three steps of sewage treatment by microbes: (1) Primary treatment (physical) — screening and sedimentation remove large solids and grit; the main component of sewage is organic matter,

which remains. (2) Secondary treatment (biological) — aerobic microbes (bacteria, protozoa) in trickling filters or activated sludge digest the dissolved organic matter, greatly reducing its quantity; the sludge is then settled. (3) Tertiary treatment (advanced) — remaining nutrients, pathogens and fine particles are removed by filtration, chlorination or UV treatment, so the treated water can be safely discharged or reused. Microbes are the key agents of the secondary (biological) stage, converting harmful organic matter into harmless substances.